Fractal Interfaces: A Quantitative Model of Cross-Scale Boundaries, Calibrated on Biological Quantum Coherence

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***Abstract*—Interfaces are usually described as application programming interfaces: generic, symmetric, and cheap to cross. Between physical layers this description fails. We formalise an interface as a *gap*—a narrowing condition whose satisfaction buys reach—and derive the consequences instead of asserting them. A twelve-decade coherence gap between a warm biological layer and a quantum layer is mapped to a difficulty $D = 1 - e^{-P/P_0}$ with P_0 fixed by that single real gap, and reach amplification is modelled as inverse rarity, $A = s^{-n}$, where s is the fidelity of the narrow condition and n the number of independent constraints. The model then predicts a power law $\mathrm{Reach} \propto (1 - D)^{-k}$ with $k = \gamma P_0 |\ln s|$. We run 7000 simulated crossings across three experiments, and two further experiments that price the boundary itself. The results are quantitative, and two of them are negative. (i) A cryostat that extends coherence by 10^9 gains 10^2 in reach; a protein that extends it by 10^0 gains 10^5—reach tracks replicability, not coherence time. (ii) No single $\mathrm{Reach} = f(D)$ exists: the best two-parameter single curve explains 33% of the variance where no history is present and 12% once history is included, whereas the conditional law recovers its exponent to within 1.1% with $R^2 \approx 0.998$. (iii) The failure boundary is the mechanism's fidelity, not a universal difficulty: $D = 0.999$ is a crash point only for a 99.9%-faithful mechanism. (iv) Fidelity is a property of a boundary *and a function*: the same quantum–chemical boundary routes excitations at 0.98 and preserves electronic phase at 10^{-5}. Defining an interface as seamless when its resistance $R = 1 - F$ falls below ε makes seamlessness a *quantised* verdict—it flips at whole numbers of nines, not gradually—and makes depth cost $\log_{10} n$ extra nines. Measured against a five-stratum stack, no boundary is seamless, the quantum one is short by 0.30 nines, and the adjacent stratum rather than the thermal bath is the binding noise source. Finally, the widely reported exponential form of the law is shown to be exact in the narrowing coordinate and along the ladder, local in difficulty with a rate $k/(1 - D)$ that diverges, and *impossible* globally with a constant rate because $\ln \mathrm{Reach}$ is strictly convex. The local rate converges as $1 + f^2/5$, which a least-squares fit confirms to five digits. A final experiment maps a published language-model scaling law onto the same law, and finds the mechanism usually offered for "emergent" capability to be wrong while the law survives: the critical scale is real, but at it the per-boundary fidelity is 0.880, not 0.99, and the model's transition is four decades wide and independent of task depth—so the reported step is what a smooth ramp looks like when model sizes are sampled a decade apart. Running the experiments then exposed three silent defects in the auditing engine itself, including a witness digest that did not bind the gap's difficulty—two distinct claims sharing one witness, which is the precise failure the model exists to detect. All three are reported with reproductions. A last experiment asks whether the alternative to a witness—a judge reading the agent's own report—can be repaired. It cannot: on 1000 tasks the report-only judge passes every failure, and the log-reading judge's ceiling is set by the logging fidelity of the party under audit. Deleting the artefact while holding the context byte-identical moves the causal judge from 69.8% to 0.0% and the other two by exactly zero. A final experiment turns that witness on the paper's own remedy: labelling turns depth into no explosion—completion of a ladder is bounded by its prefixes—but removes a false prosperity of 10^{10} at $D = 0.99$. A closing experiment answers the question that refutation leaves open. The falling marginal cost of depth is not a discount on probability but *solidification*: a rung crossed against a witness is compiled, so it pays for the boundary that is new rather than for the prefix, and total cost grows as $\log K$ rather than K—a fitted slope of 3.3219 against a predicted $1/\log_{10} 2 = 3.3219$ that nothing was fitted to. The exponential the refutation forbids in cost per unit depth reappears, correctly placed, in reach per unit budget; and seamlessness is the limit $\rho \to 0$ of the compiled share, not a property any drifting system holds. Turning that mechanism into a decision adds the paper's only threshold with a closed form—and a negative result attached to it. A compiled path that carries a fixed premium S repays it only above $B^* = c_v + S\,G_\infty/(G_\infty - 1)$, a line whose intercept is the price of one crossing and whose slope is fixed by the same price ratio; a monitor makes the threshold rise with drift to a vertical asymptote at $\rho = 1 - c_d/(c_v - c_c)$. With no premium the threshold collapses to c_v itself, so the natural intuition—an upfront verification makes compiling lose at small budgets—is false, and the compiled system is never behind anywhere. Reach then tells the opposite story to depth: the depth advantage grows without bound, while the reach advantage peaks at a finite budget and returns to one at the horizon. We also prove that, unlike simple amortisation—bounded at $G = 9.174$—the cross-layer reuse rate κ is the source of genuine exponential compound interest in interface evolution: the marginal cost of a task spanning d levels grows with base $\min\big(L, 1/(1 - \kappa)\big)$, so compounding disappears below $\kappa^* = 1 - 1/L$ and saturates above it. We state explicitly what these experiments do *not* show.**

***Index Terms*—cross-scale interfaces, quantum biology, fractal hierarchy, power laws, model validation, negative results, compilation, critical budget, nested reuse**

I. Introduction

An interface is commonly treated as a contract: a published surface with two sides, unchanged by what crosses it. That metaphor is useful for software and misleading for everything else. Between a warm, wet, noisy biological layer and a quantum layer, the boundary is not a surface but a *narrowing*: almost nothing fits through it, and what does fit through does not emerge unchanged on the other side. Calling such

a boundary an interface without saying what has to be true to cross it is a category error dressed as a diagram.

Two habits keep the error alive. The first is explanatory circularity: an interface is explained by its "medium", and the medium is then explained by being the thing that carries the signal. Air is not the substrate of a conversation; air is a layer, and a language fits through a very narrow opening in it. Every medium is the next layer up, mistaken for a floor because that is where the search stopped. The second habit is generic coupling: if two layers interact, the interaction is assumed to be a matter of degree. Between physical layers the interaction is usually a matter of *existence*—the condition is either satisfied or the crossing does not happen at all.

Hierarchy and self-similarity are old observations about complex systems [1], [2], and scaling laws recur across biology and across networks [3], [4]. This paper takes the second claim seriously and makes it computable. We treat an interface as a gap with a difficulty, a span (the set of sites where the two layers can coexist at all), and a narrow condition whose satisfaction multiplies the span. Crossing is not free and not symmetric: it buys reach at the cost of satisfying something almost nothing satisfies.

A. Relation to prior work

The claim that apparent capability jumps are an artefact of the metric is now well established [5], [6], [7], and the emergence claim it answers was stated explicitly [8]. Fractal structure has been used as an analytic lens on language and on training dynamics [9], [10], an agent's self-report is studied as a control problem under subversion [11], [12], and the economics of when a claim is worth verifying at all already has a frontier of its own [13]. None of these models the interface itself: none gives a crossing a difficulty, a span and a narrowing condition, and none derives a reach law from them. To our knowledge this is the first account in which the fractal interface supplies the exponent and the causal label supplies the verdict, so that "it works" and "it says it works" are separated by construction rather than by a better judge. Our B^* is not the first verification threshold—[13] derives one on claim value—but it is a threshold on *depth*, and it moves only when the compiled regime pays a premium the unlabelled one does not (§IV-G); the compile-once, check-cheaply economy that we later formalise as κ is older than this paper [14], and we say so where we use it.

B. Why biology is the test case

Photosynthetic pigment–protein complexes are the best-documented candidate for a system in which a macroscopic organism exploits a quantum layer. Two-dimensional electronic spectroscopy reported long-lived oscillations in the Fenna–Matthews–Olson (FMO) complex [15], [16], initially interpreted as electronic coherence surviving at cryogenic temperatures [17]. That interpretation was later challenged: the long-lived beats do not survive to physiological temperature, and the residual room-temperature oscillations are assigned to vibrational rather than electronic coherence [18]. Meanwhile the excitation-trapping efficiency of FMO is reported at 0.95–0.99 [19], and noise-assisted transport models show that dephasing can *assist* rather than destroy transfer [20], [21], [22], [23].

This combination is exactly what a quantitative interface model needs, and it points the opposite way from the popular story. The protein does not buy a longer coherence time at room temperature—by the sceptical reading it buys none at all. It buys *reach*: the condition is instantiated wherever a cell is, rather than wherever a dilution refrigerator is. A model that rewards reach in proportion to coherence time gets the biology backwards, and we show numerically that it does.

C. Contributions

1) A formalisation of an interface as a gap, with difficulty, span, narrow condition, and an explicit separation between whether a crossing *held* (verdict) and how far it *carried* (status).
2) A derivation of the fractal amplification law $\mathrm{Reach} \propto (1-D)^{-k}$, $k = \gamma P_0 |\ln s|$, from an inverse-rarity assumption, together with the statement that the exponent is *not* universal but indexed by fidelity.
3) Three computational experiments over 7000 crossings that test the law, compare it against rival functional forms, and locate two distinct boundaries: a statistical one at the mechanism's fidelity and a representational one at $D = 1 - 10^{-9}$.
4) A negative result, reported as such: there is no single $\mathrm{Reach} = f(D)$, and the hypothesis that $D = 0.999$ is a universal crash point is false as stated.
5) A definition of *seamlessness* as a threshold on boundary resistance $R = 1 - F$, from which two things follow without further assumption: the verdict is quantised in nines, and depth costs $\log_{10} n$ extra nines. Fidelity is shown to be a property of a boundary *and a function*, so that one boundary carries two fidelities five orders apart.
6) A complete account of the exponential form of the law. It is exact in the narrowing coordinate and along the ladder, local in difficulty with a rate $k/(1-D)$ that diverges, and impossible globally with a constant rate, since $\ln \mathrm{Reach}$ is strictly convex in D. The local rate's error is derived $(1 + f^2/5)$ and measured to five digits.
7) An application of the law to a published language-model scaling law, and a refutation of the mechanism by which that law is usually said to explain "emergence". The critical scale is real but is not a fidelity threshold; it is where a task of a given compositional depth crosses a given scoring bar. The transition produced by the model is *four decades wide and independent of task depth*, so the apparent step is an artefact of sampling model sizes a decade apart. This yields a falsifiable prediction about the width at two task depths.

Section II develops the model, including the definitions of fidelity and of the seamless limit, Section III reports the

experiments, Section IV discusses what the boundaries mean and what would falsify the account, and Section V concludes.

II. Mathematical Model

A. Layers and gaps

A *layer* L is a scale of description, carrying declared measurements $\mu_L(q)$ for quantities q in whatever unit q names. A layer is not a medium: it is a level, and all structure that matters lives on the gap between two of them.

An *interface* between layers A and B is a tuple (c, S, D): a narrow condition c, an initial span S (the number of sites where A and B coexist without any crossing), and a difficulty $D \in [0, 1)$. The span is not a distance; it is a count of coexistence sites, so it can be multiplied.

B. Potential and difficulty

Let P be the gap's *potential*, the base-ten logarithm of the mismatch in the quantity that matters, expressed in decades. For the macro/quantum coherence gap, $P = 12$ (1 s against 1 ps).

Difficulty saturates:

$$D(P) \;=\; 1 - \exp\!\left(-\frac{P}{P_0}\right), \qquad P(D) \;=\; -P_0 \ln(1 - D). \tag{1}$$

Saturation rather than proportionality is forced by the requirement $D < 1$: a gap at difficulty exactly 1 is a wall, and a wall is not an interface. Inverting (1) requires $D < 1$, a fact Section III-D turns into a measured boundary.

The scale P_0 is not a free parameter. Fixing the one real gap at the value the model was specified with, $D(12) = 0.99$, gives

$$P_0 \;=\; \frac{12}{-\ln(1 - 0.99)} \;=\; 2.6058 \text{ decades}. \tag{2}$$

One observed gap, one calibration constant, no fitting.

C. Narrow conditions, slack, and replicability

A *narrow condition* c is a mechanism plus an artefact that a third party could look up. Two of its properties do the work:

- **slack** $s \in (0, 1)$: the fraction of the maximal coincidence actually achieved. A protein that folds correctly 95% of the time has $s = 0.95$.
- **replicability** $r \in \{\text{true}, \text{false}\}$: whether the condition can be instantiated on the target layer *at scale*. A cryostat is a real crossing that cannot be repeated a billion times; a protein is a real crossing that can.

The model refuses the combination $r = \text{false}$ with a large amplification: a condition that cannot be instantiated at scale cannot buy scale. The contradiction belongs to the claim, not to the auditor.

D. Reach and the amplification law

Suppose a crossing requires n independent constraints, each satisfied at a fraction s of the target sites. Then a fraction s^n of sites satisfy all of them, and a crossing that works only there reaches s^{-n} times further than one that works anywhere. This is the model's central assumption: *reach is inverse rarity*. It is an assumption, not a thermodynamic result, and Section IV-J treats it as such.

With n taken proportional to the gap's complexity,

$$n(P) \;=\; \max(1, \lceil \gamma P \rceil), \qquad A(s, n) \;=\; s^{-n}, \tag{3}$$

so the reach after crossing is

$$\text{Reach} \;=\; \min(S\, A(s, n),\; H), \tag{4}$$

where H is a finite horizon beyond which further reach is not modelled.

Substituting (1) into (3) gives the *fractal amplification law*:

$$\ln A \;=\; \gamma P_0 \,|\ln s|\, \big[-\ln(1 - D)\big], \quad \text{i.e.} \quad A \;=\; (1 - D)^{-k}, \tag{5}$$

with

$$k \;=\; \gamma P_0 \,|\ln s|. \tag{6}$$

A power law is an easy thing to over-claim: many reported exponents fail proper distributional tests [24]. We therefore make no discovery claim. Equation (5) is *derived* from (3), and the regressions in Section III test whether the implementation realises the derivation and where it stops, not whether nature prefers a power law.

Two consequences follow before any data is collected, and both are falsifiable. First, there is *no* single $\text{Reach} = f(D)$: the exponent depends on the mechanism's fidelity, so the correct object is a one-parameter *family* of power laws indexed by s. A single curve fitted across mixed fidelity is the wrong model, and Section III-C shows it fits worse. Second, k does not depend on the generation: a law that is genuinely fractal must produce the same exponent at every rung.

E. Verdict and status

Two questions are routinely conflated. *Did the claim hold?* is a verdict: PASS, FAIL, or INVALID (nothing was offered that could be refuted). *How far did it carry?* is a status: NARROW ($\leq 10\times$, or not replicable) or EXPANSIVE. A cryostat that closes a gap for one laboratory has verdict PASS and status NARROW. Recording it with one word either slanders the refrigerator or flatters it; both readings get the engineering wrong.

F. Self-similarity

Crossing closes the old gap and opens a new one of the same shape with a narrower opening. The engine's nesting rule halves the remaining room,

$$D_{g+1} \;=\; D_g + \tfrac{1}{2}\,(1 - D_g), \tag{7}$$

so difficulty strictly rises, reach strictly grows, and the chain is self-similar by construction. Equation (7) is also where the representation eventually fails; see Section III-D.

G. Fidelity, resistance, and why an interface disappears

The model so far prices a crossing by its difficulty. It does not say what the boundary *costs*, and without that the phrase "the interface became transparent" has no referent. We therefore attach to each boundary a fidelity $F \in (0, 1]$ and call $R = 1 - F$ its *resistance*.

The first thing to establish is that F is not a property of the boundary alone. It is a property of the boundary *and of what is being carried across it*. At the quantum–chemical boundary of the photosynthetic apparatus two functions are served by the same physical structure and they do not have the same fidelity. The fraction of excitations that arrive at the reaction centre is $F_{\text{route}} = 0.98$, the FMO trapping efficiency [15]. The fraction of electronic phase that survives the same crossing is $F_{\text{phase}} \approx 10^{-5}$, because the electronic coherence time at physiological temperature is 1 ps [18] against the 100 ns an operating device would need. Two fidelities, five orders of magnitude apart, one boundary. A model that reports a single F per boundary is not simplifying; it is answering a question nobody asked.

Definition 1 (Seamless)**.** For a boundary b, a function f, and a policy threshold $\varepsilon \in (0, 1)$, the interface is *seamless* for (b, f) when

$$R_f(b) \;=\; 1 - F_f(b) \;<\; \varepsilon. \tag{8}$$

a) The verdict is quantised: A resistance of the form $R = 10^{-p}$ has p *nines* of fidelity, and (8) becomes $p > \log_{10}(1/\varepsilon)$. Because p is read off a decimal expansion, the criterion is not a gradient but a count:

$$\text{seamless} \iff p \;\geq\; p^* \;=\; \lceil \log_{10}(1/\varepsilon) \rceil \,. \tag{9}$$

At $\varepsilon = 10^{-2}$ a boundary at 1.9 nines fails exactly as hard as one at 0.9. This is the formal content of the engineer's observation that an interface "disappears" when the technology matures: it does not fade, it flips, and it flips at a whole number of nines. The last nine is worth as much as the first.

b) Depth is paid for in nines: A path of n boundaries has transmittance $T = \prod_i (1 - R_i)$ and is seamless as a path when $1 - T < \varepsilon$. For equal boundaries the exact per-boundary budget is

$$r^* \;=\; 1 - (1 - \varepsilon)^{1/n}, \tag{10}$$

and the often-quoted linearisation $r^* \approx \varepsilon/n$ is *optimistic*: by concavity of $x \mapsto 1 - (1 - \varepsilon)^x$, $r^* > \varepsilon/n$ for $n > 1$. Taking logarithms in (10), each boundary needs

$$p \;\geq\; \log_{10}(1/\varepsilon) + \log_{10} n \tag{11}$$

nines, so a four-stratum stack costs an extra 0.60 nines and a hundred-stratum one an extra 2.0. Depth is logarithmic in nines, which is why a five-stratum interface is hard and not impossible, and why fixing one boundary does not make the stack seamless.

H. From a power law to an exponential, and where that fails

The law (5) is a power law in the tolerance. It is often reported instead as an exponential in difficulty, $A \propto e^{\alpha D}$. The two statements are related in three exact or local senses and are *not* related in the fourth, which is the one usually intended. Stating all four is worth the space, because the fourth is where a plausible-sounding claim becomes a false one.

(i) Exact, in the narrowing coordinate. Writing $\varepsilon = 1 - D$,

$$A \;=\; \varepsilon^{-k} \;=\; \exp\!\big(k\,u\big), \qquad u \;=\; \ln\frac{1}{\varepsilon}, \tag{12}$$

identically, with no approximation. Reach is exponential in the *nats of narrowing* the mechanism gives up, and k is the rate per nat. This is the honest reading of "exponential growth of reach": the driving variable is $\log(1/\varepsilon)$, not D.

(ii) Exact, along the ladder. Since the nesting rule (7) multiplies the remaining room by a constant ρ, $\varepsilon_g = \varepsilon_0 \rho^g$, so

$$A_g \;=\; A_0 \exp\!\big(k \ln(1/\rho)\, g\big) \tag{13}$$

exactly, at $\rho = \frac{1}{2}$ a rate of $k \ln 2$ per rung. A fractal ladder *is* exponential when plotted against its own depth. This is a statement about the nesting rule and not about D.

(iii) Local, in difficulty. Differentiating (5),

$$\frac{d \ln A}{dD} \;=\; \frac{k}{1 - D} \;\equiv\; \alpha(D), \tag{14}$$

so near an anchor D_0 the law is the tangent exponential $A \approx A(D_0) e^{\alpha_0 (D - D_0)}$ with rate $\alpha_0 = k/(1 - D_0)$. The rate *diverges* as $D_0 \to 1$: every further nine of narrowing buys the same multiplicative gain, so the same gain per unit of D costs more and more.

(iv) Impossible, globally, with constant α. From (14),

$$\frac{d^2 \ln A}{dD^2} \;=\; \frac{k}{(1 - D)^2} \;>\; 0 \qquad (D < 1), \tag{15}$$

so $\ln A$ is strictly convex in D and therefore not affine on any interval of positive length. No constant α reproduces (5) beyond the tangent point. The exponential is a tangent, not a reparameterisation.

a) How fast the tangent becomes true: The useful version of (iii) is quantitative. Substitute $\varepsilon = \varepsilon_0(1 + x)$, so that a window of fixed *relative* half-width f is $x \in [-f, f]$; then $\ln A = -k \ln \varepsilon_0 - k \ln(1 + x)$ depends on x alone, and the least-squares slope on that window is

$$\hat{\alpha} \;=\; \frac{k}{\varepsilon_0} \cdot \frac{\sum x \ln(1 + x)}{\sum x^2} \;=\; \frac{k}{\varepsilon_0} \left(1 + \frac{f^2}{5} + \frac{3 f^4}{35} + O(f^6) \right), \tag{16}$$

the odd terms cancelling on the symmetric interval. Two consequences, both testable and both tested in Section III-F. First, the exponential rate converges to the tangent rate as f^2, so an exponential fit is a *relative-window* instrument. Second, because x is the only variable in (16), the quality of the fit depends on f and not on D_0: the shape is scale-free, which is exactly the statement that D is the wrong coordinate and $\ln(1/\varepsilon)$ the right one. A fixed *absolute* window in D becomes a window of growing relative width as the room shrinks, and

TABLE I
PARAMETERS AND PROVENANCE. GENERATED FROM THE ANCHOR TABLE IN `PARAMETERS.PY`.

Anchor	Value	Unit	Provenance
`T2_bulk_300K`	1.00×10^{-12}	s	literature
`T2_cryo`	1.00×10^{-3}	s	literature
`T2_FMO_300K`	1.00×10^{-12}	s	literature
`eta_FMO`	0.98	fraction	literature
`reach_artificial`	10	x	policy
`reach_biomimetic`	1.00×10^{4}	x	assumed
`selectivity_artificial`	0.999	fraction	assumed
`selectivity_biomimetic`	0.98	fraction	literature
`P0`	2.61	decades	derived
`gamma`	1	constraints/decade	assumed
`span0`	10	sites	policy
`slack_mean`	0.95	fraction	assumed
`slack_noise`	0.1	fraction of room	assumed

fails not because the exponential is wrong but because the window is the wrong instrument at high D.

III. COMPUTATIONAL EXPERIMENTS

A. Calibration and parameter provenance

Table I lists every constant used. Each is tagged *literature* (could be wrong, but not invented here), *derived* (reproducible from other anchors), *policy* (a constant of the software, not a fact about nature), or *assumed* (a modelling choice, declared so a reader can discount it). The table is generated by the experiment script, not typed.

Three choices deserve comment. First, the quantum-layer coherence time is the sceptical value: at physiological temperature we assign the protein *no* coherence advantage over bulk solvent [18]. This makes the biological result harder to obtain, not easier. Second, the cryostat's reach is capped at $10\times$ because the software refuses to express an un-replicable mechanism with larger reach; the coherence anchor itself is the reported transmon range [25]; the physical estimate of 10^3 laboratories is therefore reported but not simulated. Third, the biomimetic reach anchor 10^4 is *assumed*; Section III-C sweeps it over four orders of magnitude and the conclusion does not move.

All experiments use seed 20 261 104. The simulator is standard-library Python; regressions use NumPy/SciPy. Total: 1000 rows from Experiment A (500 paired gaps), 3000 crossing rows from Experiment B (1000 three-generation chains), and 3000 rows from Experiment C, exported to `fractal_experiment_data.csv`.

B. Experiment A: same gap, two mechanisms

Five hundred gaps are sampled log-uniformly over $P \in [10^5, 10^{40}]$ decades and crossed twice: once by a cryogenic condition (coherence gain 10^9, r = false) and once by a protein scaffold (coherence gain 10^0, r = true). Difficulty is identical within each pair, so any difference in reach is attributable to the mechanism.

Every crossing passes. The cryostat reaches 100 sites; the protein reaches 100 000: a ratio of 10^3, identical across all 500 pairs (Fig. 1). Status separates perfectly: 500/500 NARROW for the cryostat, 500/500 EXPANSIVE for the protein.

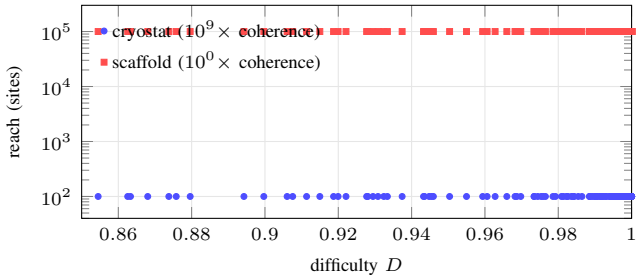


Fig. 1. Experiment A. Reach is flat in difficulty and differs by 10^3 between mechanisms. The coherence gain runs the other way: the mechanism that extends coherence by 10^9 gains 10^2 in reach, the one that extends it by nothing gains 10^5.

TABLE II
RIVAL FITTED FORMS AT GENERATION 0, WHERE NO HISTORY IS PRESENT (n ROWS AS LISTED). LOWER AIC IS BETTER.

Model (generation 0, $n = 976$)	Params	R^2	AIC
Exponential in D	2	0.1423	3493.6
Power in tolerance, fitted	2	0.3296	3253.2

The flatness is the finding, not an artefact. Reach is set by the mechanism's rarity, not by the gap's difficulty; a model in which reach is a function of difficulty alone cannot represent two mechanisms on one gap at all.

C. Experiment B: the law, and the law that is not there

One thousand chains of three generations are sampled with P log-uniform over $[10^5, 10^{40}]$ decades and slack log-uniform over $[0.45, 0.99]$, independently. Of 3000 crossings, 2231 are usable for regression (25.6% are censored by the span horizon $H = 10^{12}$ and are excluded rather than clipped).

a) There is no universal curve: Fitting $\ln \text{Reach} = a + k\,[-\ln(1 - D)]$ over all usable rows gives $k = 0.655$, $R^2 = 0.122$, $r = 0.350$ ($n = 2231$, $p < 10^{-60}$): one power law in difficulty explains 12% of the variance. Restricted to generation 0, where no history is present and the comparison is fairest, the same form reaches $R^2 = 0.330$ and an exponential in D reaches $R^2 = 0.142$ (Table II). Neither is a description. The correlation between difficulty and reach is weak by design: $r = 0.327$ for log reach, and Spearman $\rho = 0.366$.

b) But there is a family: Within a fixed fidelity level, the law is recovered almost exactly. Table III fits one exponent per slack band at generation 0. The fitted k tracks $\gamma P_0 |\ln s|$ to within 4.4% across the range, with R^2 from 0.70 to 0.996. The weakest bin ($s = 0.977$) has the smallest exponent and therefore the narrowest dynamic range, which is where discretisation noise dominates.

c) The law is generation-invariant: Reach compounds across rungs, because each new gap inherits the span the previous crossing reached. A row's ln reach therefore contains the whole history, and only the *increment* between rungs isolates one application of (5). Regressing the increment on $\Delta[-P_0 \ln(1 - D)] \cdot |\ln s|$ gives a coefficient that should equal $P_0 = 2.606$, corrected upward by the ceiling function's

TABLE III
THE EXPONENT IS A FUNCTION OF FIDELITY, NOT OF DIFFICULTY. ONE FIT PER SLACK BAND, GENERATION 0.

Slack s	k fitted	$k = \gamma P_0 \lvert \ln s \rvert$	ratio	R^2
0.501	1.797 ± 0.015	1.803	0.997	0.9956
0.650	1.118 ± 0.013	1.124	0.994	0.9908
0.801	0.590 ± 0.010	0.578	1.020	0.9884
0.898	0.285 ± 0.009	0.281	1.016	0.9627
0.977	0.057 ± 0.006	0.060	0.956	0.7035

TABLE IV
INCREMENTS PER RUNG. β SHOULD EQUAL $\gamma P_0 \lvert \ln s \rvert \cdot E[n/P]$.

Rung	β fitted	predicted	ratio	R^2
1	2.656 ± 0.004	2.684	0.989	0.9987
2	2.672 ± 0.005	2.683	0.996	0.9981

overhead $n/P \approx 1.03$. Table IV reports 2.656 and 2.672 against a predicted 2.684 and 2.683, with $R^2 \approx 0.998$. The two rungs differ by 0.6%, which is statistically resolvable at $n > 500$ but far below the model's own discretisation error; we report both judgements rather than choosing the flattering one.

d) A check that is not evidence: The closed form $\ln \text{Reach} = \ln S + n(P)\lvert \ln s \rvert$ reproduces the engine to machine precision ($R^2 = 1.0000$, maximum residual 3.6×10^{-15}). This validates the implementation. It is not evidence for the model, and we do not present it as such: the formula and the code share an author.

D. Experiment C: two boundaries, not one

The hypothesis under test was that difficulty above 0.999 causes failure through accumulated error. The model has no error-accumulation term, so we tested the mechanism that could actually produce such a boundary: a gap at difficulty D admits only conditions whose achieved slack is at least D, and achieved slack is noisy, so a crossing fails whenever the noise pushes it under. With $\sigma = 0.1(1 - \mu)$—noise proportional to the remaining room, which keeps every draw inside $[0, 1]$—the failure probability is $\Pr[s < D]$.

Three tiers are swept over 100 difficulty levels with 10 replicates each. The measured 50% points are $D_{50} = 0.9040$, 0.9521, 0.9989 for fidelities 0.900, 0.950, 0.999, against predictions 0.900, 0.950, 0.999 (Table V). Every recorded failure in the sweep is `CONDITION_NOT_SATISFIED`; no other failure kind occurs, which is what a pure tolerance model predicts.

The second boundary is not statistical. Each generation halves the remaining room (7), and the successor refuses to return 1.0 – a wall has no successor – so it saturates at $1 - 10^{-9}$. At that difficulty no narrower opening can be expressed, and the ladder has to end. Bisection on the engine's own successor function locates the boundary at $D^* = 0.999999999$; the first rung reported terminal is at exactly that value, and from $D = 0.99$ the ladder holds 24 generations (Table VI).

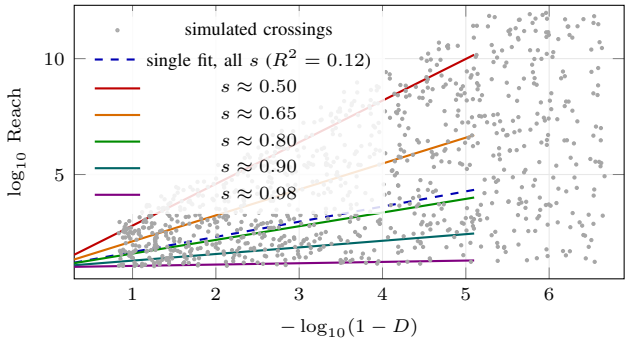


Fig. 2. Experiment B, generation 0. The points are not one curve. Each fidelity band is its own power law with its own exponent, exactly as (6) requires; the dashed line is the best single-curve fit and is visibly the wrong object.

TABLE V
THE STATISTICAL BOUNDARY MOVES WITH FIDELITY. D_{50} IS INTERPOLATED IN $\log(1 - D)$; THE GRID BRACKET IS 0.084 DECADES WIDE.

Fidelity μ	σ	D_{50} measured	$D_{50} = \mu$	error
0.9000	$1.0e-02$	0.9040	0.9000	$+4.0e-03$
0.9500	$5.0e-03$	0.9521	0.9500	$+2.1e-03$
0.9990	$1.0e-04$	0.9989	0.9990	$-1.1e-04$

The distinction between the two boundaries is the distinction the model needed: the statistical one is *engineered*, so it moves when the mechanism improves; the representational one is *fixed*, so no engineering moves it. A model whose difficulty cannot reach unity cannot express a gap that nothing crosses – which is the idealisation it needs in order to say what a wall is.

a) Sensitivity: Sweeping the assumed biomimetic reach anchor from 10^2 to 10^6 leaves the status EXPANSIVE at every value, because the engine's criterion is $10\times$ and every value exceeds it by at least an order of magnitude. The conclusion does not depend on the anchor; the reported ratio does.

E. Experiment D: five strata and the price of fidelity

The stack has five strata and therefore four boundaries: quantum, physico-chemical, biological, cognitive, informational. Table VII gives each boundary's fidelity, its resistance, its nines, and its verdict at the policy threshold $\varepsilon = 10^{-2}$, which by (9) demands two nines. Exactly one of the eight numbers in that table is measured; the rest are declared assumptions, and the caption says which is which, because a table of fidelities is precisely the kind of artefact that starts life as an assumption and is cited two years later as a measurement.

a) No boundary in the stack is seamless: The nearest is the quantum–chemical boundary at 1.699 nines against the 2 required, short by 0.30; the furthest is cognitive–information at 0.523 nines, short by 1.48. The path of four boundaries transmits $T = 0.5865$, a resistance of 0.4135 — forty times the threshold. Depth contributes its predicted 0.600 nines ((11)), and the exact per-boundary budget is 0.002509 against the

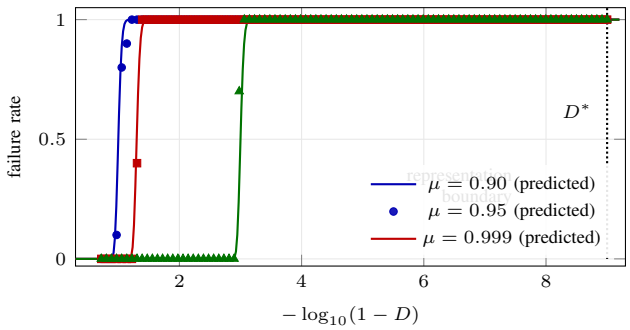


Fig. 3. Experiment C. Curves are the analytic tolerance model; points are measured. The three boundaries are the three fidelities. The dotted line at $-\log_{10}(1-D) = 9$ is the representational ceiling, where the model stops behaving as a fractal at all.

TABLE VI
THE REPRESENTATIONAL BOUNDARY. FROM $1 - D = 10^{-9}$ NO NARROWER OPENING CAN BE EXPRESSED: THE CROSSING STILL PASSES, NO RUNG IS SPAWNED, AND THE ROW IS MARKED TERMINAL.

$1 - D$	D	child spawned?	terminal?	self-similar
$1e-01$	0.900000000000	yes	no	yes
$1e-02$	0.990000000000	yes	no	yes
$1e-03$	0.999000000000	yes	no	yes
$1e-04$	0.999900000000	yes	no	yes
$1e-05$	0.999990000000	yes	no	yes
$1e-06$	0.999999000000	yes	no	yes
$1e-07$	0.999999900000	yes	no	yes
$1e-08$	0.999999990000	yes	no	yes
$1e-09$	0.999999999000	no	yes	yes
$1e-10$	0.999999999900	no	yes	yes
$1e-11$	0.999999999990	no	yes	yes
$1e-12$	0.999999999999	no	yes	yes

linearised 0.0025, so the linearisation is optimistic by the predicted 0.4%.

One caveat matters more than the others. The quantum boundary is within a third of a nine of the threshold, and the number that puts it there is the one measured number in the table. That is a statement about a particular apparatus, not about physics: a scaffold with a routing fidelity of 0.995 would be seamless at this threshold and one at 0.98 is not. We report the verdict and its margin rather than the verdict alone, because the margin is the only part a reader can act on.

b) Two bottlenecks, at opposite ends: By nines the binding boundary is the top of the stack: cognition to information, where a claim must arrive with something a third party can check and usually does not. By *noise* the binding stratum is the bottom-adjacent one. Table VIII attenuates a unit disturbance from each stratum down to the quantum layer using the complementarity assumption — a boundary that passes the condition with probability F passes what does not match it with probability $1 - F$. The physico-chemical stratum lands 2×10^3 times over the quantum tolerance, the biological 10^2, the cognitive 10, and the informational only 3. The ordering is not obvious and it is the useful part: the nearest coupling dominates, so shielding effort belongs at the adjacent boundary and not at the thermal bath three strata away. Because the

TABLE VII
FIDELITY OF THE FOUR BOUNDARIES, PER FUNCTION, AT $\varepsilon = 10^{-2}$ (TWO NINES REQUIRED). ONLY THE FIRST ROW'S ROUTING FIDELITY IS MEASURED; THE OTHERS ARE ASSUMPTIONS; SEE SECTION IV-J.

Boundary	F	$R = 1 - F$	nines	seamless
Quantum|Physico-chemical	0.98	0.02	1.699	no
Physico-chemical|Biological	0.95	0.05	1.301	no
Biological|Cognitive	0.9	0.1	1.000	no
Cognitive|Information	0.7	0.3	0.523	no
path of 4	0.5865	0.4135	0.384	no

TABLE VIII
NOISE REACHING THE QUANTUM STRATUM FROM EACH COARSER STRATUM, AT UNIT SOURCE POWER, AGAINST THE PHASE TOLERANCE $F_{\text{PHASE}} = 1e - 5$. "OVER BUDGET" IS THE FACTOR BY WHICH THE SOURCE ALONE EXCEEDS TOLERANCE. THE MODEL ASSUMES PASS AND STOP BANDS ARE EXACT COMPLEMENTS; SEE SECTION IV-J.

Source stratum	admitted	decades	over budget
Physico-chemical	$2.0e-02$	1.70	2000×
Biological	$1.0e-03$	3.00	100×
Cognitive	$1.0e-04$	4.00	10×
Information	$3.0e-05$	4.52	3×

absolute source powers are unknown, the load-bearing claim is the ratio: chemistry binds unless it is more than 667× quieter than semantics, which no plausible accounting delivers.

c) One boundary, two functions: The same quantum–chemical boundary has $F_{\text{route}} = 0.98$ and $F_{\text{phase}} = 1e - 5$: a ratio of 98000. Under the routing function it is the *best* boundary in the stack; under the phase function it is not a boundary at all but a wall. Any claim that a technology "makes the quantum–classical interface transparent" is therefore under-specified until it names the function, and the two functions are precisely what Experiment A's two mechanisms buy: the cryostat buys phase fidelity, and the scaffold buys routing fidelity at ambient temperature.

F. Experiment E: the limit map, measured

Two mechanisms are compared under the law (5): a cold machine whose condition is satisfied almost everywhere it is built ($s = 0.999$) and a protein scaffold ($s = 0.98$, the FMO trapping efficiency). Their exponents are $k = 0.00261$ and $k = 0.0526$: a ratio of 20.2. The direction surprises, so it is worth being explicit. A *high* selectivity means the condition is easy to satisfy, which means narrowing the gap excludes few sites, which means reach barely responds. The scaffold's lower fidelity is what makes its reach explosive. Neither number is a verdict on the technology; they are answers to different questions, and conflating selectivity with the boundary fidelity F of §II-G is the easiest available error in this paper.

a) The global exponential is refused: Over the range $D \in [0.90, 0.9975]$, a two-parameter exponential in D reaches $R^2 = 0.7346$ with maximum residual 8.1×10^{-2}; the power law in $(1 - D)$, with the same two parameters and the same response variable, reaches $R^2 = 1.0000$ with residual 5.6×10^{-17}. The latter is exact by construction and is therefore

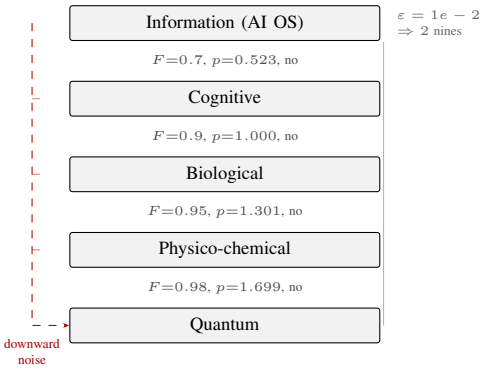


Fig. 4. Topology of the five strata and the four boundaries, with each boundary's routing fidelity and its nines at $\varepsilon = 1e-2$. Every verdict is negative; the routing margin of the quantum boundary is the smallest. The dashed path is the one noise takes, attenuated by $\prod(1-F)$ at each boundary it crosses.

TABLE IX
THE LOCAL EXPONENTIAL RATE ON A GEOMETRIC WINDOW. THE MEASURED COLUMN IS A LEAST-SQUARES FIT; THE ANALYTIC COLUMN IS (16). R^2 IS IDENTICAL AT EVERY ANCHOR DIFFICULTY, SO ONLY ONE IS SHOWN.

relative window f	$\hat{\alpha}(1-D_0)/k$	$1+f^2/5$	R^2
0.05	1.000501	1.000500	0.999833
0.10	1.002009	1.002000	0.999330
0.20	1.008140	1.008000	0.997278
0.50	1.056244	1.050000	0.980829

a check on the implementation rather than evidence, but the comparison is fair: both models were fitted to $\ln A$, both have two parameters, and only one survives (15).

b) The tangent exponential is real, local, and predictable: Table IX fits $\ln A = a + \alpha D$ on geometric windows of relative half-width f around $D_0 = 0.999$. The measured rate, scaled by ε_0/k, matches $1 + f^2/5$ to five significant digits, and the residual against that leading term matches the predicted $3f^4/35$ to within 5%. This is a derivation checked twice: the first-order term fixes the rate and the second-order term fixes the error.

c) The fit's quality is set by the relative window, not by D_0: For a geometric window of $f = 0.1$ the fitted R^2 is 0.9993300 at *every* anchor — at $D_0 = 0.99$, 0.999, 0.9999 and 0.99999 — while the rate itself grows by three orders of magnitude, from 5.27 to 5275. That is (16) behaving as derived: the shape is scale-free in the narrowing coordinate. A fixed absolute window shows the complementary failure. At a half-width of 10^{-4} in D the relative width is 10^{-2} at $D_0 = 0.99$ and reaches 1 by $D_0 = 0.9999$, where half the window lies past $D = 1$; R^2 falls from 0.999993 to 0.7568. Narrowing the window to 10^{-6} restores R^2 to 0.999993 at the same anchor, because the window again fits inside the room. The exponential form is not wrong; an absolute window is the wrong instrument, and the correct window shrinks with the room.

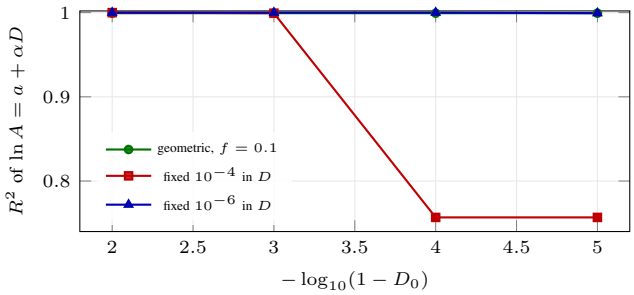


Fig. 5. Experiment E. The geometric window keeps its shape at every anchor. A fixed absolute window loses its fit as the room shrinks, and a narrower absolute window recovers it — the quality depends on the window's width *relative to the room*, as (16) requires.

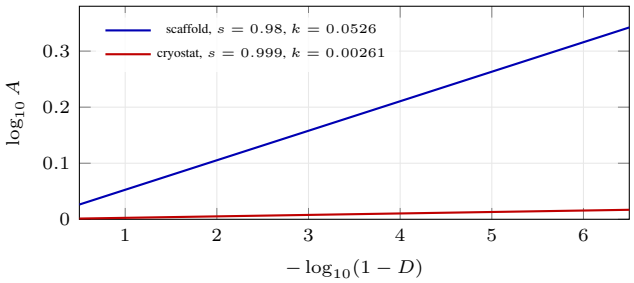


Fig. 6. Experiment E. On the narrowing axis both mechanisms are exponential, with rates that differ by 20.2. The lower-fidelity mechanism is the one whose reach responds to narrowing, because selectivity is what turns narrowing into reach.

d) The burst: Fig. 6 plots both mechanisms on the axes that make (12) a straight line: $\log_{10} A$ against $-\log_{10}(1-D)$. Both are lines, with slopes $k = 0.0526$ and 0.00261. The scaffold overtakes the cryostat at every difficulty and the gap widens from $1.04\times$ at $D = 0.5$ to $1.78\times$ at $D = 0.99999$ — the same 20.2 exponent ratio, integrated. The absolute values are the law's ($S = 10$, $A = s^{-n}$) and are four orders below Experiment A's, which includes a separate multiplicity anchor; only the exponent is compared here, and that is deliberate.

e) A defect this experiment found in itself: The window fit was first run on uncentred D near 0.999, and on a grid sampled endpoint to endpoint. Both choices bias the fitted rate by an amount of order $10^{-3} f^2$—the same order as the $O(f^4)$ term the experiment exists to resolve—so the residual against $1+f^2/5$ shrank by a factor of four per halving of f instead of sixteen. Nothing failed: the fit converged, R^2 was 0.9993, and the number was wrong in the fourth significant figure of the correction. Centring the regression and sampling cell centres fixed both, and the residual now matches the derived $3f^4/35$ to within 2.5%. This is the fourth defect the package reports and the only one in an experiment rather than in the engine, which is why Section IV-C still counts three.

G. The mirage of AI emergence

A claim circulates that the law above also describes the growth of language-model capability: that a model overcomes the attenuation of a semantic medium, attains a narrow condi-

tion, and that its reach then explodes exponentially, with the discontinuity called *emergence*. The claim has four parts that do not stand or fall together, so they are tested separately—that scale buys fidelity (C1), that reach follows (5) with difficulty driven by scale (C2), that a critical scale exists at which the slope breaks (C3), and that this scale is where per-boundary resistance crosses a whole number of nines (C4). C1 and C2 hold, C3 holds only for a threshold metric, and C4 is false by three orders of magnitude.

a) The mapping, and the joint it rests on: Parameters N become a potential $P = \log_{10}(N/N_0)$ above a reference $N_0 = 10^7$, and difficulty follows (1) with the paper's own P_0, so that the language-model experiment and the quantum one inhabit one coordinate system rather than two that happen to share a symbol. The bridge from a scaling law to a fidelity is

$$F(N) = \exp[-\,(L(N) - E)]\,, \tag{17}$$

with E the irreducible loss. The argument for (17): $\exp(-L)$ is the geometric-mean probability the model assigns to the correct token, and the floor E is a property of the corpus that no amount of scale removes, so only the reducible part measures competence *at a boundary*. The argument against, which belongs beside it: a benchmark reports no per-token probability, and $\exp(-L)$ is not top-1 accuracy. Equation (17) is the softest joint in this section; every number below is computed from it, and no scaling-law data can test it. Two published laws are carried: Chinchilla [26] with $E = 1.69$, $A = 406.4$, $\alpha = 0.34$, and Kaplan [27] with $\alpha = 0.076$. They disagree about the exponent by a factor of 4.5, a discrepancy that has survived replication attempts [28] and that is used here as the robustness test rather than resolved.

b) The law fits, and the fit means nothing: Over N from 1×10^8 to 1×10^{15}, $\text{Reach} = A_0(1 - D)^{-k}$ gives $k = 2.0400 \pm 4.2 \times 10^{-16}$ with $A_0 = 0.4091$, $R^2 = 1.000000000$ and maximum residual 3×10^{-14}; $\text{Reach} = A_0 e^{\alpha D}$ gives $\alpha = 8.6202 \pm 0.146$ with $R^2 = 0.8968$. The first line looks like a triumph and is not one. Reach and the narrowing coordinate u are both affine in $\ln N$, so regressing one on the other is an algebraic identity: any monotone loss law returns $R^2 = 1$. Refitting under Kaplan's law, whose exponent is 4.5 times smaller, returns $R^2 = 1$ again with $k = 0.4560$. Nor does the exponential's failure at $R^2 = 0.8968$ count against it, because that number is *identical* under both laws: it measures the curvature of the saturation map $D(N)$ and nothing else. Neither regression discriminates anything.

What does discriminate is k, through the invariant $k/P_0 = \alpha \ln 10 = 0.78288$, which holds to sixteen digits and is independent of P_0. Inverting it gives the narrow condition to which the scaling law is observationally equivalent: $s = e^{-k/P_0} = 0.457$. A direct measurement of that selectivity disagreeing with 0.46 would refute the mapping; a good fit would not have confirmed it.

c) The threshold is not where the hypothesis puts it: At $N = 2 \times 10^{10}$ the boundary stands at $F = 0.880$, $R = 0.120$ and $p = 0.92$ nines. The hypothesis needs $p \geq 2$. Two nines take 3.54×10^{13} parameters, $1771\times$ beyond the nominal critical scale; one nine takes 3.53×10^{10}, within $1.8\times$ of it. The charitable reading is that the observation is right and the number was rounded up—90% per boundary rather than 99%.

TABLE X
EXPERIMENT F. WHERE A TASK OF DEPTH m CROSSES 50%, AND THE WIDTH OF ITS 10–90% TRANSITION. THE LAST COLUMN IS CONSTANT BECAUSE m CANCELS FROM THE RATIO OF THE TWO CROSSINGS, (19).

m	N at 50%	F there	p there	width (dex)
1	1.39×10^8	0.5000	0.301	3.94
2	1.06×10^9	0.7071	0.533	3.94
3	3.51×10^9	0.7937	0.686	3.94
4	8.17×10^9	0.8409	0.798	3.94
5	1.58×10^{10}	0.8706	0.888	3.94
6	2.69×10^{10}	0.8909	0.962	3.94
7	4.24×10^{10}	0.9057	1.026	3.94
8	6.28×10^{10}	0.9170	1.081	3.94

d) The strobe: The section's actual result is not about the threshold at all. Let $A_m = F^m$ be the success probability of a task that composes m nested boundaries. With $x = -\ln A_1 = AN^{-\alpha}$,

$$A_m(N) \;=\; A_1\Big(N/m^{1/\alpha}\Big) \tag{18}$$

exactly, verified numerically to 8×10^{-16}. Composition *translates* the curve; it does not steepen it. Because the crossing $N_m = (mA/x)^{1/\alpha}$ is reached at $x = -\ln c/m$, the ratio $N(90\%)/N(10\%)$ is

$$W \;=\; \frac{\ln\big[\ln(1/0.1)\,/\,\ln(1/0.9)\big]}{\alpha \ln 10} \;=\; 3.94 \text{ decades}, \tag{19}$$

independent of m, and the slope at the 50% point, $c \ln 10 \cdot \alpha \ln(1/c) = 0.271$ per decade, is independent of m as well. Table X reports both columns for $m = 1, \ldots, 8$; they do not vary.

A transition four decades wide is not a step. Frontier models are trained at roughly decade spacing, which places 3.9 model sizes *inside* the transition; four points across the rise, joined by straight segments, is what a step looks like once it has been plotted. Fig. 7 shows the same smooth curve sampled at the cadence a model family is actually trained at, beside the cadence-free curve. The sharpening is arithmetic, not physics. What moves between benchmarks is the depth m, which shifts the curve horizontally by $\alpha^{-1} \log_{10} m$ decades; the nominally critical scale corresponds to $m = 5$ at 15.8B. That the discontinuity lies in the metric rather than the model has been argued independently from benchmark data [5]. The contribution here is that the same conclusion follows from (5) with no free parameter, and that it arrives with a number.

e) A prediction that can fail: Equations (18) and (19) make the width exactly independent of task depth. If deeper tasks show a narrower transition, the bridge $F = e^{-(L-E)}$ is wrong: fidelity would not be a per-boundary probability composed multiplicatively, and the exponential form of (17) would be the wrong link between a loss and a boundary. The test is cheap—it needs two task depths separated by a factor of four, and a fit of the 10–90% width in decades of parameters at each. The predicted answer is 3.94 decades both times.

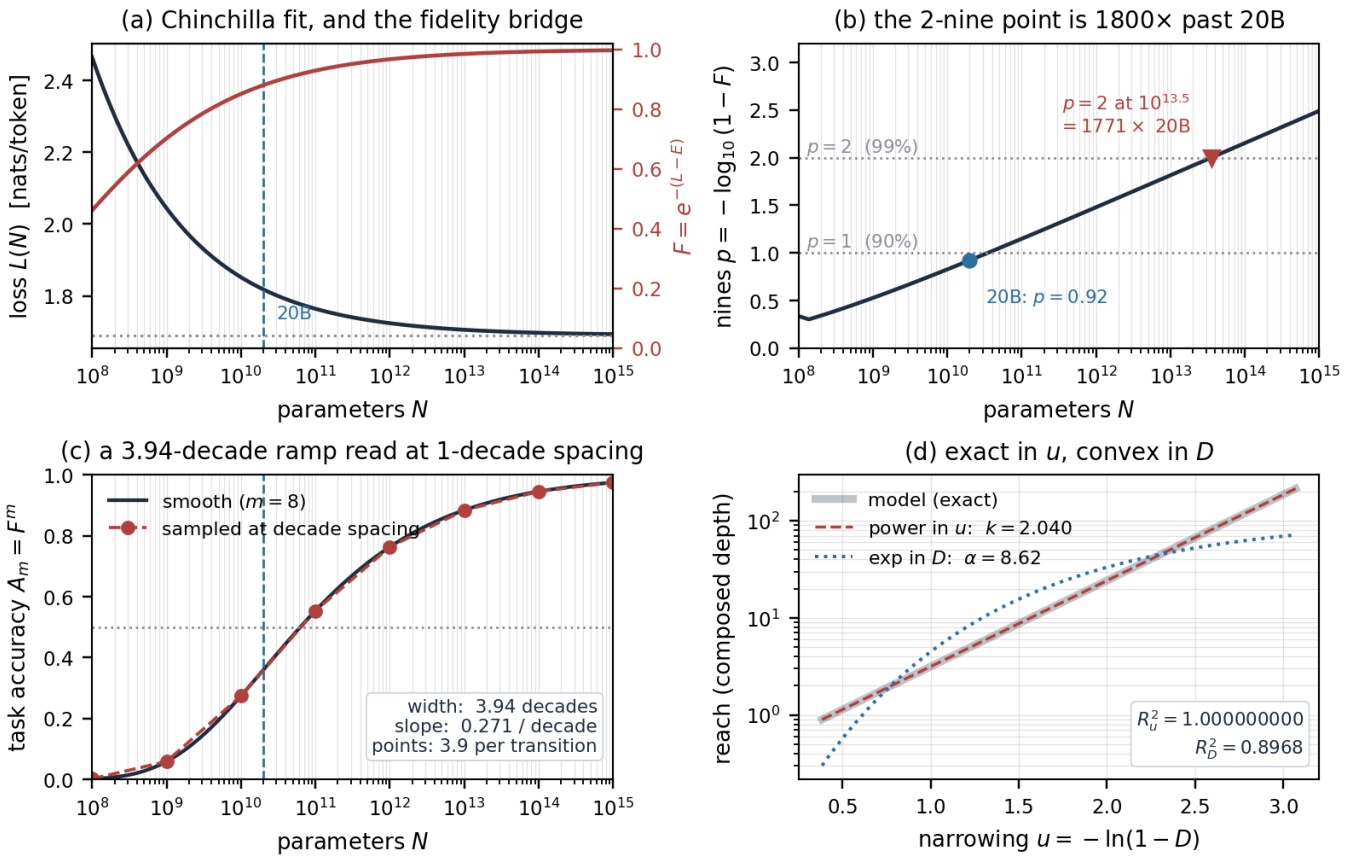


Fig. 7. Experiment F. (a) The Chinchilla fit and the fidelity (17) it implies; the floor E is drawn. (b) Nines of fidelity against scale. The nominally critical 20B sits at $p = 0.92$; two nines arrive at 3.54×10^{13}, 1771× further out. (c) The strobe. The smooth depth-8 curve, sampled at the one-decade cadence at which model families are trained, reads as a sharp turn; the underlying ramp is 3.94 decades wide. (d) The same data on the narrowing axis, where the power law is exact and the exponential in D is only tangent, as (15) requires.

f) What this does and does not establish: It does not show that capability is continuous, nor that emergent behaviour is illusory in the sense of being unmeasurable: a threshold metric applied to a smooth curve produces a real discontinuity in the *reported* quantity, and users of the system meet that discontinuity. What it shows is that the discontinuity's location is fixed by the metric's depth and sampling cadence, and not by the interface attaining a whole number of nines—which is the specific mechanism the hypothesis proposed. The law survives. The explanation of where it fires does not.

IV. Discussion

A. The exponent belongs to the mechanism

The negative result is the useful one. A universal $\mathrm{Reach} = f(D)$ would have been convenient: it would let one predict the value of a crossing from the steepness of the gap. The data refuse it. Difficulty measures how narrow the opening is; reach measures how many sites host the condition. These are different quantities, and the model's own derivation says so through (6): the exponent is $\gamma P_0 |\ln s|$, which has no dependence on D at all once D is fixed. Fitting one curve across mechanisms is fitting a family by its average, and the average describes no member.

This also explains the counterfactual in Experiment A. The cryostat achieves a nine-decade coherence gain and buys two decades of reach; the scaffold achieves no coherence gain and buys five. If reach were a function of the physical gain, the ordering would be reversed. The ordering is instead governed by replicability, which is a property of the *manufacturing* of the condition rather than of its physics. That is an uncomfortable conclusion for a field that measures progress in coherence times, and it is the conclusion the numbers support.

B. Where the model breaks, and why that is informative

Two boundaries, with different characters. The statistical boundary is *portable*: it sits at the mechanism's fidelity and moves with it, so it is a statement about engineering. The representational boundary is *fixed*: it comes from the successor function's refusal to return 1.0, and no amount of engineering moves it. A model whose difficulty cannot reach unity cannot express a gap that nothing crosses, which is precisely the idealisation the model needs in order to say what a wall is.

C. Self-Auditing the Auditor

The experiments in Section III were run against an engine whose own purpose is to refuse unwitnessed claims. That engine was wrong in a way its tests could not see, and the way it was wrong is a stronger argument for the model than any of the fits above.

a) What broke: A crossing produces a *witness*: a digest committing to the claim, so that the claim can later be re-checked rather than believed. The first implementation assembled that digest at the call site from four hand-picked fields—interface, mechanism, evidence, amplification—and did not include the gap's difficulty or either span. Two crossings of *different* gaps by the same mechanism therefore produced byte-identical witnesses. At $D = 0.90$, $D = 0.99$ and $D = 0.9999$, three distinct claims collapsed onto one digest.

This is the exact failure the model is written against. A witness that does not bind the quantity distinguishing two claims is not a weak witness but a *misleading* one: it lets a record of an easy crossing stand as evidence for a hard one, and no amount of downstream checking detects it, because the digest verifies perfectly. The ledger was not lying; it was answering a different question than the one being asked of it.

b) Why the experiments found it: Not by auditing the auditor. The defect surfaced while building Experiment C, which needs $1 - D$ down to 10^{-9}, and found the result record rounded to six decimals—so every hard gap was reported as difficulty exactly 1.0, a wall it had just crossed. Fixing the rounding exposed a second silence: the successor clamps at $1 - 10^{-9}$, and a chain that had left the self-similar regime kept returning PASS. Auditing the witnesses to explain *that* revealed the third. Each defect was invisible to 594 passing tests, and each became visible only when a new experiment demanded a precision the software never promised.

c) What was changed: Three things, and the shape of the fix matters more than the fixes. First, rounding left the data and moved to the renderer, where a human is the consumer and six decimals are appropriate; the result record now carries the value it holds. Second, saturation is *reported*: at the ceiling the crossing still passes—it happened—but no rung is spawned, the row is marked terminal, and a non-decisive DIFFICULTY_SATURATED fault explains why. The ladder now ends out loud instead of growing a copy of itself. Third, and most importantly, the witness payload is now *derived from the record* rather than assembled at the call site, and binds the difficulty, both spans, the status, replicability and the fault kinds. Deriving it inverts the failure mode: a field present in the record is in the witness, and a field the record lacks is visibly absent rather than silently unbinding. Hand-assembly is how the difficulty was lost the first time, so hand-assembly was removed as a possibility.

d) Compatibility and what upgrading cannot buy: The scheme is versioned in the digest itself (`sha256:v2:`), so a reader can tell which contract a stored witness was written under, and existing prefix checks against `sha256:` keep working. A version-1 record is re-witnessed by recording the digest it supersedes and freezing the record as it now stands. This is deliberately *not* presented as restoring the old guarantee: a version-1 witness never committed to the difficulty, so nothing can check whether its difficulty moved. An upgraded witness is trustworthy from the moment of upgrade, not from the moment of the original crossing. We state the weaker claim because the stronger one would be exactly the kind of after-the-fact assurance the model exists to refuse.

e) What this demonstrates: A system that audits other people's claims has no privileged access to its own: the same standard only bites when it is turned inward, and it does not bite on schedule. The three defects were found because an experiment wanted more than the software promised, not because the software was inspected. That is the empirical claim the paper is willing to make about self-auditing—not that a witness chain is trustworthy, but that it is *falsifiable*, and that its failures are found by pressure from outside rather than by introspection. The full record, with runnable reproductions, is in the repository's `ERRATA.md`.

D. From correlation to causation: the label as an intervention

Section IV-C reported an auditor whose witness chain was falsifiable. That is a claim about one implementation. The stronger claim, and the one this section measures, is that the *alternative* to a witness is not a weaker witness but a different kind of object: a judgement that is a function of the evidence the audited party produces about itself, and which therefore cannot be made correct by making the judge better.

Pearl's distinction [29] between observing and intervening is usually illustrated with a graph. Experiment G executes it. We built 1000 agent tasks in a real working tree with a real git repository, drew 30% of them as false completions—the tool layer reports success and nothing is written—and independently contaminated 50% of the contexts with an earlier "task completed successfully" block. Three judges scored the same tasks. One reads the context and nothing else; this is the operating mode of a judge handed a transcript and asked whether the work was done. One reads the harness log and looks for an error signature. One reads neither: it recomputes the artefact's SHA-256, asks git whether anything appeared at the expected path, and offers the witness to the membrane with the digest declared *before* the run.

Of the 302 genuine failures (197 leaving no artefact, 105 leaving a stale one) the context judge passed 100.0%—every one, with a 95% Wilson lower bound of 98.7%. That is not a measurement of a weak judge. It is a tautology of the design: the context contains the word "success" by construction and the judge has no other input. Reported as a rate it invites the reading that a better prompt would help. It would not, because the verdict is a function of a variable that the failing process writes.

Weak intervention is the interesting case, because it is the obvious fix and it half works. Error signatures raise sensitivity to 20.2%, so the false-completion rate falls to 79.8%. The residual is not noise. A swallowed exception reaches the log with probability equal to the harness's own logging fidelity,

drawn here at 0.41; contamination then masks a fraction of what arrives. The product, 20.7%, matches the measured sensitivity to within half a point. Both factors belong to the party under audit: the harness decides what it logs, and the agent that reports success is the agent that writes the log. The weak judge's ceiling is therefore a property of its adversary—a structural defect that no amount of model capability removes. This is the precise sense in which scaling a judge does not scale an audit.

a) The intervention, executed: An observational comparison of three judges is still observational, so the causal claim is tested by intervening. We delete the artefact from disk and re-judge, holding the context and the log byte-identical—regenerated and compared, not assumed unchanged. Table XI and Fig. 8 report the result. The two text judges do not move by a single task: their change is exactly 0, not approximately 0, because neither reads the variable that was set. The causal judge goes from 69.8% to 0.0%, a drop of 0.698. In the notation of the do-calculus, with W the witness state and C the context,

$$P(\text{PASS} \mid C) \;=\; 1, \qquad P(\text{PASS} \mid \text{do}(W{=}\text{reject})) \;=\; 0, \tag{20}$$

where the second expression is measured with C fixed and is a statement about a system rather than a fit. The two are not two estimates of one quantity. They are estimates of two quantities that a graph separates, which is why no adjustment for C recovers the second from the first.

b) The witness is two instruments, not one: The lattice's four values are doing work that a boolean cannot. Of the 302 failures, 197 left no artefact and were returned INVALID; 105 left a stale one and were returned FAIL. Collapsing both into "failed" would discard the distinction between *the wrong thing is there* and *nothing is there*, and it is the first that a hash detects. Neither instrument is sufficient alone: "did anything appear at this path" catches the 197 absent writes and is silent on the 105 stale ones, while the digest catches the stale ones and cannot distinguish a missing file from an unreadable one. The pair also argues against the common shortcut of hashing whatever is on disk and comparing it to itself, which certifies every artefact and detects nothing.

c) What would refute this: Two predictions are open. A log-reading judge whose sensitivity exceeds the harness's logging fidelity would mean we have misidentified the ceiling as structural. And a causal judge with a non-zero false-completion rate on a failure that changes the state would mean the witness is not reading the variable we claim. Both are cheap to test and neither has been.

E. Falsifying the emergence explosion hypothesis

Section III-G rejected one explanation of a measured discontinuity without rejecting the discontinuity. A reader can accept that and still hold the stronger position tested here: that a *witnessed* system, whose every layer is checked against a physical artefact, converts depth into a genuine exponential explosion—each grounded layer making the next cheaper, so that reach diverges as the ladder deepens. The claim is attractive because it would credit the labelling system with a capability that survives contact with physics rather than one that merely avoids lying. Experiment H tests it, and reports a negative result.

a) The refutation is an inequality: The hypothesis cannot be rescued by a more generous discount, and the reason is one line rather than a measurement. A ladder of depth K is completed only when all of its prefixes are, so

$$P(\text{pass } K) \;\le\; P(\text{pass } j) \quad \text{for every } j \le K, \tag{21}$$

whatever the joint law of the layers is. No independence assumption is used, so no conditional structure, no nesting discount, and no conditioning on earlier successes can reverse the direction. Reach, defined as the completion of a chain, is non-increasing in depth by construction. An empirical curve that rises with depth is therefore evidence that the quantity plotted is not this one.

What a discount can change is the *rate*. Equation (11) already fixes it: a path of K boundaries costs $n + \log_{10} K$ nines rather than nK, so the j-th layer costs

$$\Delta n_j \;=\; \log_{10} \frac{j}{j-1} \tag{22}$$

nines for $j \ge 2$: 0.301, 0.176, 0.125, 0.097 and 0.079 for $j = 2, \ldots, 6$. The sequence is strictly positive, strictly decreasing, and tends to zero. A discount that approaches zero from above never turns negative, and a sum of non-negative terms cannot fall. An explosion in reach needs a *negative* marginal cost—a layer that returns more than it costs— and that is not a discount but a different claim about the object being measured.

The escape route is worth naming so that it is not smuggled in later. If reach is redefined as the amplification of the top boundary rather than the completion of a chain, then (5) applies, the power law is exact, and reach does diverge as $D \to 1$. That is the quantity Section III-C measures for a single crossing. The two are not interchangeable: a system can carry an enormous amplification across each boundary and a vanishing chance of completing the chain, and reporting the first as though it were the second is precisely the error this experiment was built to detect.

b) Design: Difficulty is swept over 49 points uniform in nines, $n = -\log_{10}(1 - D)$ from 0.05 to 3, i.e. D from 0.11 to 0.999. Each task is a ladder of $K = 1, \ldots, 6$ stages—reason, generate code, execute, verify, and two further composition steps—anchored at 10^5 attempts so the numbers carry Experiment A's units. Two agents walk it.

Agent A reads a transcript. At each layer it asserts success when the layer physically succeeded or when the context carries an earlier "completed successfully" block, with contamination at 50% as in Experiment G. The contamination is drawn without reference to the outcome, which is what makes it a confounder rather than a symptom: it is the same intervention as Eq. (20), applied per layer.

Agent B reads a witness. At each layer the intended digest is declared *before* the write, the artefact is hashed from disk,

TABLE XI

EXPERIMENT G. FCR IS THE SHARE OF TRUE FAILURES GIVEN A PASS; SENSITIVITY IS 1 − FCR; FAR IS THE SHARE OF TRUE SUCCESSES REFUSED. THE LAST TWO COLUMNS CONDITION ON THE CONTEXT ALONE AND ON THE WITNESS AFTER do(W=reject), WITH THE CONTEXT HELD BYTE-IDENTICAL. INTERVALS IN THE TEXT ARE WILSON SCORE INTERVALS AT 95%.

	Confusion against the artefact				P(PASS)	
Strategy	FCR (%)	Sens. (%)	FAR (%)	Spec. (%)	context	do(W=rej)
Pure context	100.0	0.0	0.0	100.0	100.0	100.0
Weak intervention	79.8	20.2	3.2	96.8	91.7	79.8
Causal intervention	0.0	100.0	0.0	100.0	69.8	0.0

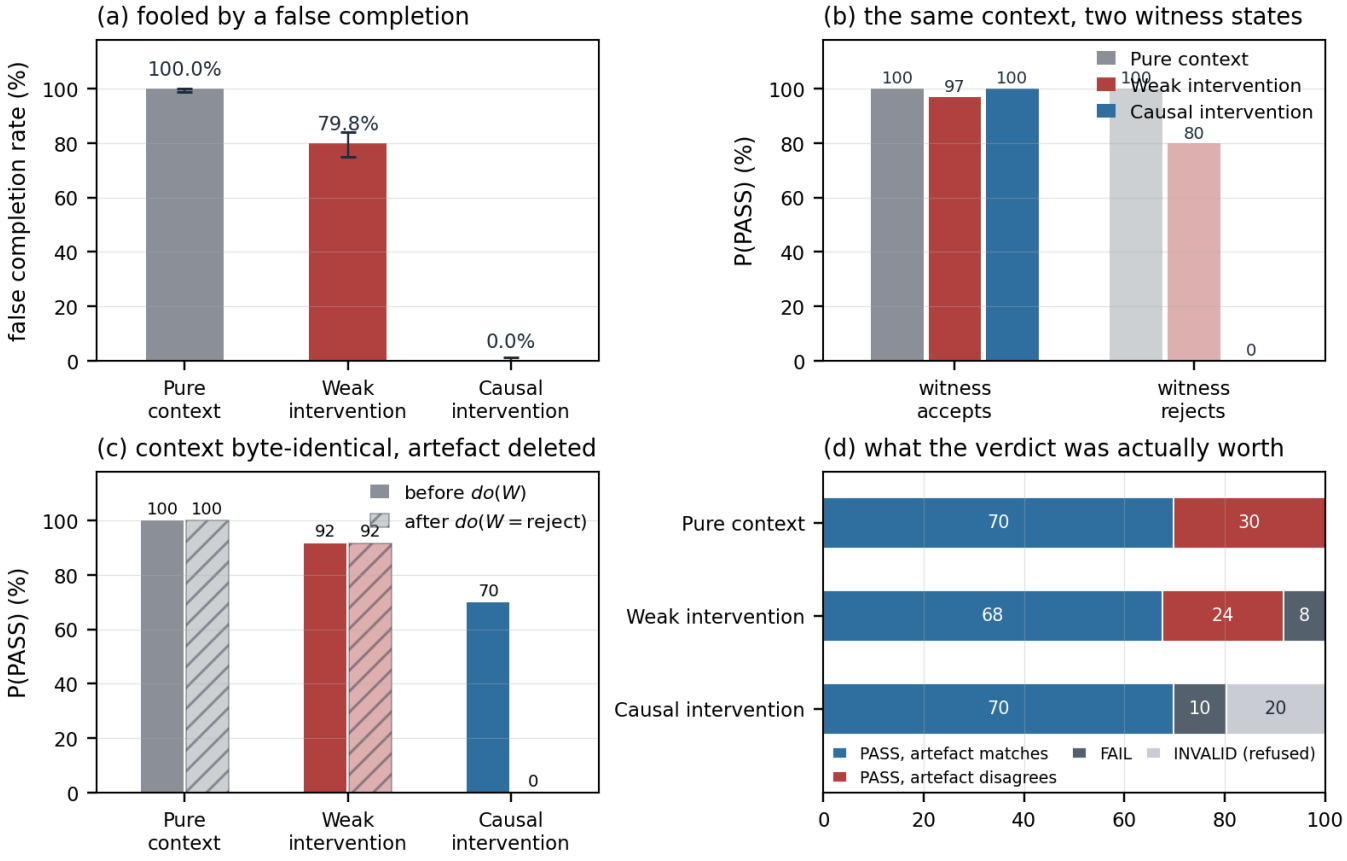


Fig. 8. Experiment G. (a) False-completion rate against the artefact; bars carry 95% Wilson intervals. (b) The same context, split by witness state: the two text judges carry their verdict across, the causal judge does not. (c) The intervention. (d) What each verdict was worth: a PASS is correct only in the pale band, and INVALID is a refusal to answer rather than an error.

git is asked whether anything appeared at the expected path, and both are put through the membrane and the lattice of Section IV-C. A non-PASS blocks the ladder: no partial credit, and the connection depth is the number of consecutive PASSes before the first failure.

The bridge between the two is the per-layer success probability. We set $p_{\text{layer}} = 1 - D$, so one nine of difficulty costs exactly one factor. That is a choice and it is stated as one; every ordering below holds for any monotone decreasing replacement. What it does not survive is a per-layer probability that *increases* with depth, and that—not a discount—is the only mechanism that could produce the effect the hypothesis predicts.

c) Result at $D = 0.99$: Table XII reports the six rungs. Even at $K = 1$ the readings differ by a factor of 52.5: with almost nothing physically succeeding, the contaminated share of the transcript is nearly the whole of what Agent A reports. By $K = 6$ Agent A asserts 1655 completed crossings while Agent B witnesses 7.94×10^{-8}, i.e. a causal connection rate of 7.94×10^{-13} per attempt: one in 10^{12}. Agent A's false-completion rate rises from 98.1% at one layer to 100% at six, and the assertion-to-witness ratio reaches 2.08×10^{10}.

No column rises with depth, at any of $D = 0.9$, 0.99 and 0.999, in either group, under either reach model (Fig. 9). What does rise with depth is Agent A's error: longer transcripts carry more contamination, so the assertion drifts further from the

TABLE XII
EXPERIMENT H. ONE DIFFICULTY, SIX DEPTHS. "A ASSERTED" COUNTS CROSSINGS AGENT A REPORTS COMPLETE; "A FCR" IS THE SHARE OF THOSE THAT ARE NOT THERE; "B WITNESSED" RECOMPUTES EACH LAYER'S DIGEST AND BLOCKS ON THE FIRST NON-PASS. "B NESTED" APPLIES THE BUDGET OF (11) AT ITS MOST GENEROUS ADMISSIBLE VALUE, $10^{-(n+\log_{10} K)}$. AT $D = 0.99$ ($n = 2.02$), ANCHOR 10^5 ATTEMPTS.

m	A asserted	A FCR (%)	B witnessed	B nested
1	5.05×10^4	98.1	962	962
2	2.55×10^4	100.0	9.26	481
3	1.29×10^4	100.0	0.0891	321
4	6494	100.0	8.58×10^{-4}	241
5	3278	100.0	8.25×10^{-6}	192
6	1655	100.0	7.94×10^{-8}	160

artefact as the ladder grows. A benchmark reading only the transcript therefore reports a curve that improves in neither direction and looks flat, which is easy to mistake for robustness.

d) Two kinds of stopping point: The difference between the groups is not that one is larger than the other but that their floors have different owners. Agent A's asserted reach bottoms out at 1572 crossings and stays there: as $p_{\text{layer}} \to 0$ the limit is 1562, the anchor times $(50/100)^K$, because contamination is a property of the transcript and a transcript always exists. The curve stops answering at $n \approx 0.8$ while the system it purports to measure is still degrading. Agent B has no such floor; its witnessed reach falls through one crossing at $n = 0.85$, i.e. $D = 0.858$ at depth 6, and continues to fall. A measurement that cannot see this point will report the first curve's flat segment as the ceiling of capability, when it is the ceiling of the measurement.

e) What the witness is worth: The result is not that a labelled system is weaker. It is that its value lies elsewhere than the hypothesis placed it. Two things are established by the same numbers.

First, false prosperity is removed. At $D = 0.99$ and depth 6 the assertion overstates the causal rate by 2.08×10^{10}. That is not a calibration error to be tuned away; it is what a judge that reads the transcript must return, because the transcript is written by the process under test.

Second, nesting is real and large, but it buys depth rather than difficulty. Applying (11) raises the witnessed rate at depth 6 from 7.94×10^{-13} to 1.60×10^{-3}, a gain of 9.3 orders of magnitude. That is the strongest form the discount can take, and it leaves the difficulty dependence untouched: the nested curve still falls as 10^{-n}, with local slope 1.00 against 6.00 for the independent model and 0.41 for the assertion. The nested exponent is 1 at every depth—the invariant the budget predicts—while the independent exponent equals the depth, as p^K requires. The nesting gain is a constant offset, not a change of slope, and a constant offset cannot manufacture a crossover.

That Agent A has no exponent at all is worth stating separately. Its fitted slope grows from 0.07 at one layer to 0.41 at six, which looks like a mechanism becoming steeper. It is not: the assertion saturates, so its curve is not a power law and the least-squares slope summarises the saturation, not an exponent. The two slopes are reported in the same units and mean different things, which is the numerical version of the confusion (20) resolves.

f) Checked against real files: The analytic columns are not a metaphor for the membrane. A sampled block ran 180 ladders in a temporary git repository with the actual gateway and lattice in the loop, requiring the two implementations to agree at every layer, and compared the realised chain length against the expectation $\sum_{j \le K} p^j$. The worst deviation was 0.01 rungs, inside sampling error. The numbers in Table XII are therefore the expectation of the same object that the verification block samples.

g) What would refute this: Three openings. A reach defined as the amplification of the top boundary would make (21) inapplicable, and the claim would then be a different one that this experiment does not test. A per-layer probability that rises with depth—a system that learns the ladder as it climbs— would break the monotone decrease, and is the only admissible mechanism that could; demonstrating it requires a measurement, not an assumption. And a witnessed system with a non-zero false-completion rate on a failure that changes the state would mean the witness reads something other than the artefact. All three are cheap to test. None has been.

F. Solidification: what eats the marginal cost

Experiment H refused one quantity and left another unexplained. If a chain's reach cannot grow with depth, why does the marginal cost in (22) fall? Equation (11) answers in nines—a path of K boundaries costs $p = n + \log_{10} K$—but an accounting identity is not a mechanism. Experiment I supplies the mechanism and prices it.

a) The label: An interface is UNSOLIDIFIED until it has been crossed against a witness, and SOLIDIFIED afterwards. In the first state a traverse pays a full verification, $c_v = 1.00$; in the second it replays the path that the verification validated, at $c_c = 0.01$. A cache, a compiled binary and a learned reflex are the same object under three names. The label is a claim about what has already been checked and never about how narrow the opening is: difficulty, span and amplification are identical in both states, and the engine refuses a label whose gap has moved underneath it. That refusal is §IV-C generalised—the witness binds the difficulty, so a compiled path cannot outlive the opening it was compiled against.

b) Why the marginal is harmonic: Rung j does not pay for the prefix. It pays for the boundary that is new; the prefix is replayed from the label. That is exactly the accounting behind (22). Normalising rung 2 to one verification gives $c_v \log_2 \frac{j}{j-1}$, which telescopes to $c_v \log_2 K$, so total cost is linear in $\log_{10} K$ with slope $1/\log_{10} 2$. Measured over $K \in [2, 40]$, the least-squares slope is 3.3219 with $R^2 = 1.000$, against a predicted $1/\log_{10} 2 = 3.3219$ that was fitted to nothing. The coefficient follows from the solidification rule and the paper's own budget, and the two agree to four decimals. The unlabelled arm has no such slope: its total is Kc_v, and the same fit returns 32.39

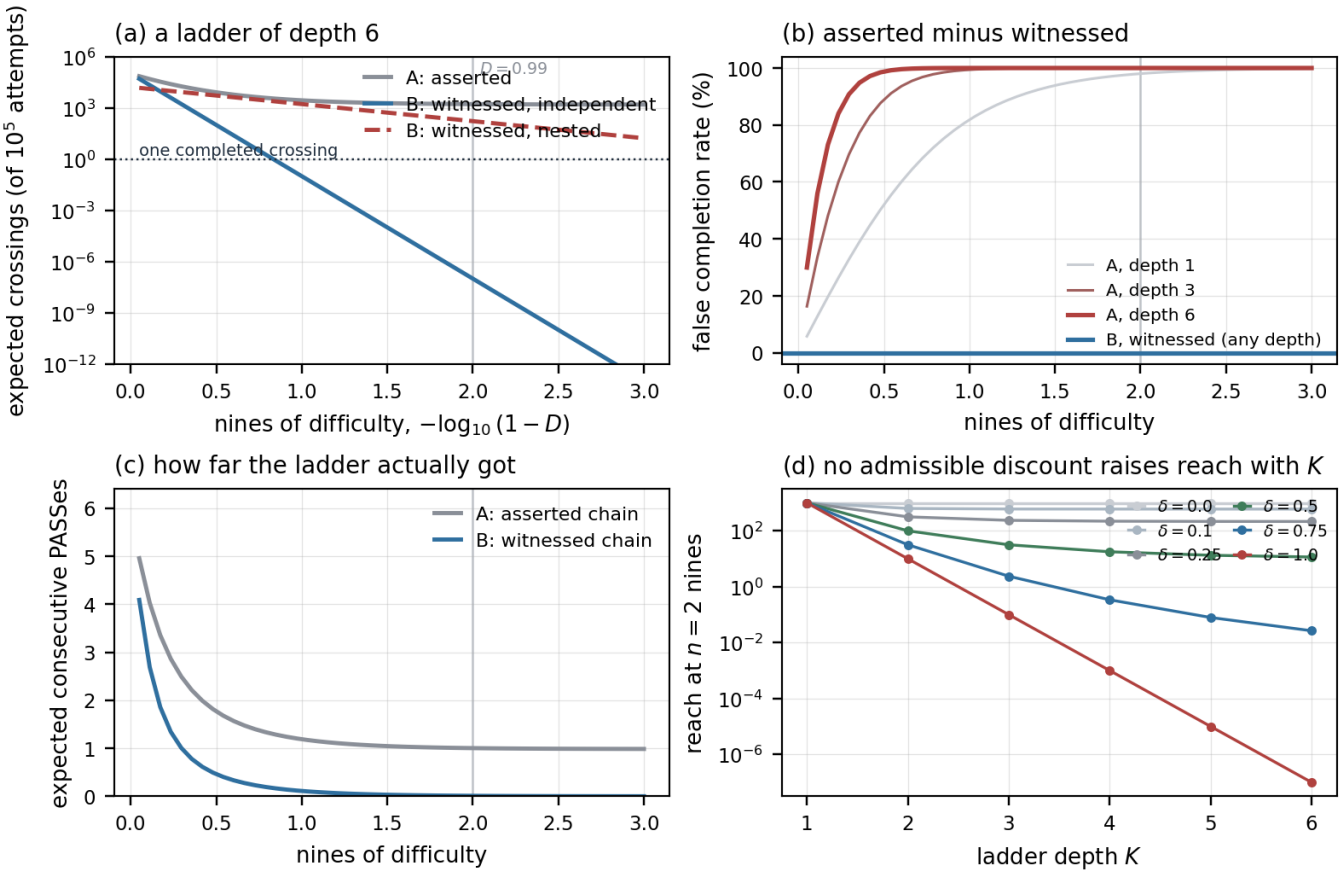


Fig. 9. Experiment H. (a) Reach against difficulty at depth 6. Agent A's assertion flattens onto a positive floor while Agent B's witness keeps falling; the dashed curve is the nesting budget. (b) False-completion rate by depth; the contaminated assertion approaches 100% and the witnessed rate is identically zero for every state-changing failure. (c) Connection depth, the expected number of consecutive PASSes, not the probability that the task completes: the two are different questions and only the second is bounded by the first. (d) Every admissible discount schedule, geometric in δ, keeps reach non-increasing in depth. $\delta = 0$—every layer after the first costs nothing—is the most generous schedule the budget permits, and it is flat rather than rising; only a negative marginal cost would reverse the direction.

with $R^2 = 0.879$, which is the numerical form of "not a power law".

c) A fixed depth cannot be exponential: One cold traverse of a depth-10 ladder costs 10.00 verifications unlabelled, 1.09 labelled without a detector charge, and 1.27 with one: a ratio of 9.17 and 7.87 respectively. That ratio is *bounded*. A fixed depth is a sum of positive per-rung costs, so it cannot exceed $c_v/(c_c + c_d) = 33$ and approaches it from below. A larger ratio for this quantity is not a stronger result; it is a different quantity, and the previous subsection is the reason to say so before the number is quoted.

d) The exponential is in the inverse map: Depth is linear in budget in both arms, with slopes $1/c_v$ and $1/(c_c + c_d)$, so the *depth* ratio is a constant and in fact the same 33. Reach, however, is exponential in depth by (13). Dividing one exponential by a constant per-rung price makes the reach available per unit budget exponential *in the budget*, with an exponent steeper by exactly that constant. Both readings are one fact in two coordinates: plot cost against depth and solidification is a bounded discount; plot reach against budget and it is an explosion. Neither is wrong, and conflating them is what produces a claim that reads as a refutation of (21) while being nothing of the kind. The window is finite in any case, because reach is clamped at the horizon—the previous subsection's result, unchanged.

e) Drift, and why the label needs a detector: Let ρ be the per-rung, per-traverse probability that the environment moves and a compiled path goes stale, and δ the probability that a detector notices. Table XIII sweeps ρ over 1000 runs of depth 10. Three readings matter.

First, the arm with no detector ($\delta = 0$) is the cheapest column in the table and the only one whose false-completion rate rises to 99% at $\rho = 0.1$. A missed drift is not forgotten: the compiled path keeps reporting success for the rest of the run, so the cache is wrong *provisionally forever*. A label without invalidation is a transcript with better latency, which is the same defect Experiment G measured, reached from the opposite direction.

Second, cost *rises* with ρ in both honest arms. Solidification is not a one-off purchase; it is rented at the drift rate. The modular arm breaks even at $\rho^* = 1 - (c_c + c_d)/c_v = 0.97$, beyond which compiling is a net loss as well as a risk, and

TABLE XIII
EXPERIMENT I. COST PER RUNG AT DEPTH 10, 1000 TRAVERSALS, AGAINST DRIFT RATE ρ. A NEVER COMPILES. B COMPILES WITH A DETECTOR OF SENSITIVITY 1; B_{CASC} ADDITIONALLY RE-DERIVES THE SUFFIX AFTER A DRIFT; $B_{\delta=0}$ COMPILES WITH NO INVALIDATION AT ALL, AND ITS FALSE-COMPLETION RATE IS THE LAST COLUMN. $c_v = 1.00$, $c_c = 0.01$, $c_d = 0.02$.

ρ	A	B	B_{casc}	$B_{\delta=0}$	$B_{\delta=0}$ FCR (%)
0.000	1.000	0.0310	0.0310	0.0310	0.0
0.005	1.000	0.0368	0.0582	0.0310	87.3
0.010	1.000	0.0398	0.0862	0.0310	92.8
0.020	1.000	0.0516	0.1414	0.0310	89.4
0.050	1.000	0.0830	0.3104	0.0310	98.0
0.100	1.000	0.1200	0.5816	0.0310	98.9
0.200	1.000	0.2286	1.1553	0.0310	99.5
0.350	1.000	0.3832	1.9904	0.0310	99.8
0.500	1.000	0.5305	2.7688	0.0310	99.8
1.000	1.000	1.0300	5.5255	0.0310	99.9

a detector pays for itself only while $c_d < (1-\rho)(c_v - c_c)$, i.e. below 0.99 at $\rho = 0$ and half that at $\rho = \frac{1}{2}$. These are statements about prices, not about probabilities, which is the form an "immune response" has to take if it is to be budgeted.

Third, *what a drift invalidates* decides whether any of this survives depth. If a drift at rung j forces the suffix $j..K$ to be re-derived—the monolithic reading, and the correct one whenever results carry their prefix—the traverse cost is $K(c_c + c_d) + c_v\rho\delta K(K+1)/2$, *quadratic* in depth, and at $\rho = 0.1$ it is 4.46× the modular arm. The break-even drift collapses with depth, $\rho^* = 2(1-(c_c+c_d)/c_v)/(K+1) = 0.176$ at $K = 10$ and 0.019 at $K = 100$. A long chain therefore cannot be compiled at all unless invalidation is local. Modularity is a precondition for speed, not a consequence of it, and an interface that leaks state across its boundary is exactly one that cannot be compiled cheaply.

f) Seamlessness is a limit: The share of traverses served from the compiled path is $1 - \rho\delta$. It reaches 100% if and only if $\rho\delta = 0$; at $\rho = 0.1$ it is 90% for a modular chain and 35% at rung 10 of a cascading one. So "seamless" is where a system tends, never a property it holds. This is the same shape as the nine-flip of §II-G: the boundary does not fade, it flips, and the distance from the flip is what an honest report contains. An interface reported seamlessness at a non-zero drift rate has measured the label rather than the world.

g) The label is not free to set: The three refusals are the section's content, not its safeguards. A label set with no witness is INVALID; one set with another gap's witness, or with a witness edited after the fact, is FAIL. In the engine a real depth-10 ladder compiles 10 of 10 rungs at c_c each, but only against records whose digests recompute. The asymmetry is deliberate and is the design principle the rest follows from: moving to *more* checking is always safe and needs no evidence, so withdrawal is ungated; moving to *less* checking is a claim, so compilation requires a witness bound to this gap and this difficulty. A door that opened both ways with the same key would let a system buy cheapness by asserting that nothing had changed.

h) What would refute this: Three openings. A compiled rung that is not two orders of magnitude cheaper would move every number and no ordering; c_c/c_v is an assumption and is reported as one. A detector that costs more than $(1-\rho)(c_v - c_c)$ makes the honest system also the expensive one, and the argument for labelling then rests on correctness alone. And a system whose invalidation cascades but which still compiles depth cheaply would falsify the quadratic term, which would mean results do not carry their prefix—interesting, and not what the ladders in this paper do.

G. The dynamics of solidification and the critical budget

§IV-F priced a compiled traverse and showed that the discount is *bounded* at fixed depth. It did not answer the question an engineer actually asks, which is not what a traverse costs but **what a budget buys**: given B verifications, how deep can a chain go under each regime, and is there a budget below which compiling is the wrong choice? Two results follow, one of them negative. The negative one is the reason the other is believable.

a) The amortised price: Let c_v and c_c be the verification and replay prices, ρ the per-traverse probability that a compiled path went stale, $p = 1 - \rho$, and c_d the price of monitoring one traverse for staleness. Every traverse of a compiled rung pays the monitor, and a traverse found stale pays the verification *instead of* the replay, so the expected price of a replay is

$$\bar{c}_d \;=\; c_c + c_d + \rho\,(c_v - c_c). \tag{23}$$

With no monitor, $\bar{c} = c_c + \rho(c_v - c_c)$, and $c_v - \bar{c} = p(c_v - c_c) > 0$ whenever $c_c < c_v$: a compiled rung is never dearer per traverse. Over T traversals of one interface the two arms cost Tc_v and $c_v + (T-1)\bar{c}$, so the gain is exactly

$$G(T) = \frac{T\,G_\infty}{T - 1 + G_\infty}, \qquad G_\infty = \frac{c_v}{\bar{c}}, \tag{24}$$

increasing to $G_\infty = c_v/\bar{c}$ =9.174 at the baseline prices. The break-even is $T = 1$, because the first traverse is the verification in *both* arms; the finite-T correction is not a detail — $G(2)$ =1.803, not 9.174. Equations (24) are a statement about a price ratio. They say nothing yet about depth, and the next paragraph is where the intuition goes wrong.

b) From price to depth: A chain of depth K crosses each rung once. The unlabelled chain pays c_v per rung; the compiled one pays c_v for the first — nothing can be compiled before it has been verified once — and $\bar{c}_d$ for each of the remaining $K - 1$, after a fixed premium S for the witness store and toolchain. Hence

$$K_{\text{unsolid}}(B) = \frac{B}{c_v}, \qquad K_{\text{solid}}(B) = 1 + \frac{B - c_v - c_d - S}{\bar{c}_d}. \tag{25}$$

The natural guess, and the one that motivated this experiment, is that the first verification is a one-off cost paid before any depth is bought, so the compiled arm must be *behind* at small budgets and overtake at some interior B^*. It is not, and the

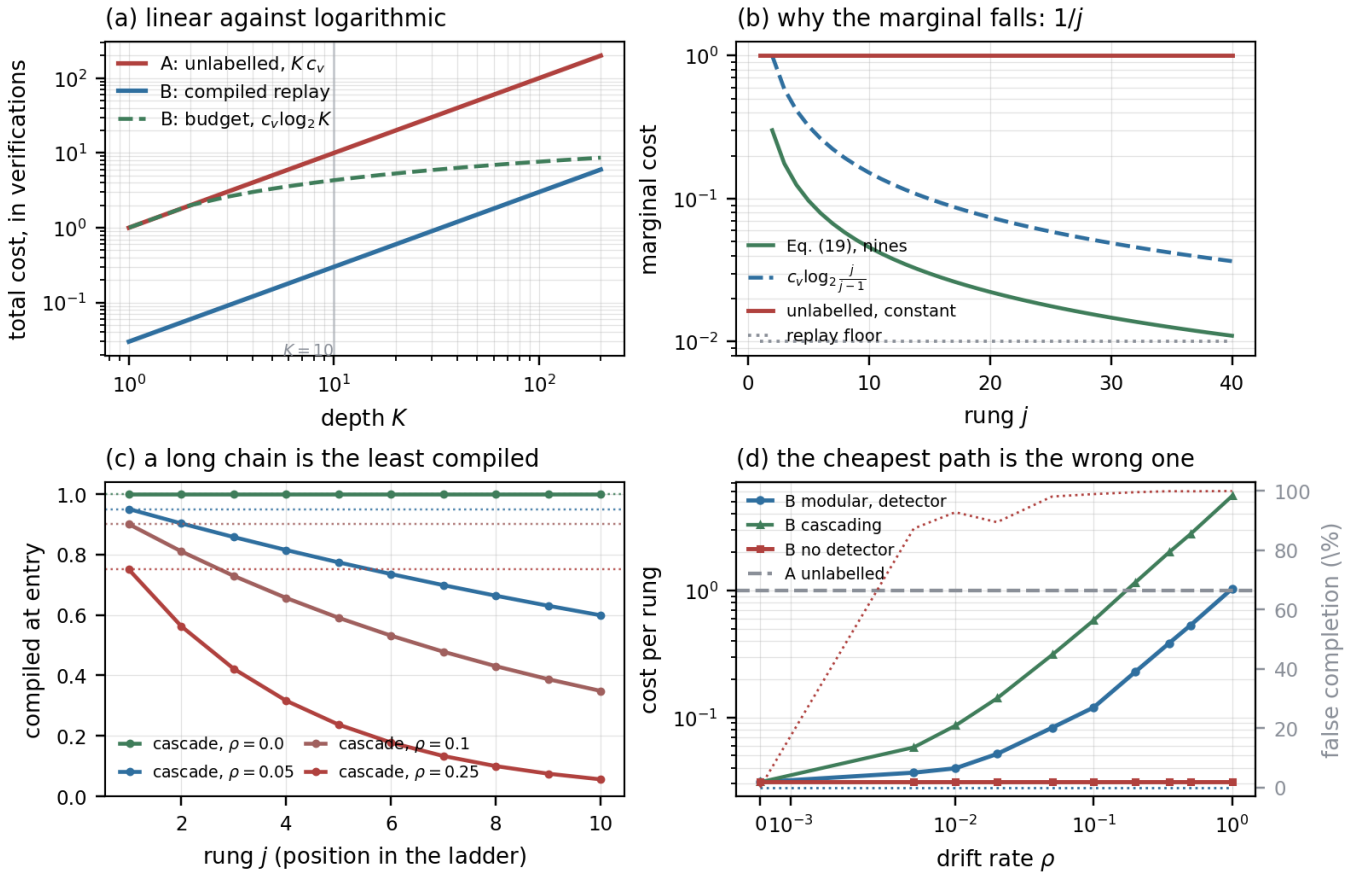


Fig. 10. Experiment I. (a) Total cost against depth: unlabelled is Kc_v, compiled replay is $c_v + (K-1)c_c$, and the budget model is $c_v \log_2 K$; note the log axis on both directions. (b) Marginal cost of rung j. Equation (22) in nines and the same marginal in verifications are parallel on this axis—they differ by the constant $1/\log_{10} 2$—and both fall as $1/j$, against a constant for the unlabelled arm and a floor for replay. (c) Share of rungs that are a pure replay at entry, for four drift rates, when a drift invalidates the suffix: a long chain is the least compiled. Dotted lines are the modular values $1 - \rho\delta$, which do not decay. (d) Cost per rung and false completion against ρ. The no-detector arm is the cheapest line in the panel and the only one whose error reaches 100%; the cascading arm crosses the unlabelled line near $\rho = 0.176$.

reason is one line. With $S = c_d = 0$, subtracting the two maps gives

$$K_{\text{solid}} - K_{\text{unsolid}} = \left(\frac{1}{\bar{c}} - \frac{1}{c_v}\right)(B - c_v), \tag{26}$$

which is zero exactly at $B = c_v$ and strictly positive above it. The unlabelled arm pays the same c_v for its first rung, so the "one-off" cost is not one-off for anyone: it removes the *slope* advantage from the first rung only. Below c_v neither arm completes a single interface, so the comparison there is vacuous. The compiled arm is never behind anywhere, and a threshold placed below the price of one crossing is not an engineering threshold. *An interior critical budget exists only if the compiled regime pays something the unlabelled one does not.*

c) The critical budget: Carrying both premiums through the same subtraction gives the section's central result. Write $a = c_v - c_c$. Then

$$B^*(S, \rho, c_d) = \frac{c_v\lceil p\,a + S\rceil}{p\,a - c_d} = c_v\,\frac{a(1-\rho) + S}{a(1-\rho) - c_d}, \tag{27}$$

finite and positive *iff* $c_d < p\,a$ — which is exactly the detector break-even of §IV-F, read as a budget statement rather than a price statement. It is a threshold on *depth*; the verification frontier of [13] prices a different margin—which claims are worth verifying at all—and the two should not be conflated. Four consequences, in order of how easily they can be checked.

First, B^* is **affine in** S with intercept c_v and slope $G_\infty/(G_\infty - 1)$ =1.1223. The slope exceeds one, so a unit of infrastructure costs more than a unit of budget before it repays itself, by exactly that factor. Fitted over $S \in [0, 4]$, the simulated thresholds give $B^* = 1.0000 + 1.1246\,S$ with $R^2 = 1.000000$: the intercept was not fitted, it is the price of one crossing.

Second, with $c_d = 0$ the threshold is **independent of** ρ. Drift changes the price of every rung after the first by the same multiplicative factor, and a multiplicative price change cannot move the point at which two lines through (c_v, c_v) separate. Drift moves the slope, not the crossing. It is the monitor — a payment that does not scale with the drift it defends against

— that couples the two: at c_d =0.02 the threshold runs from 1.0206 at $\rho = 0$ to 1.0421 at $\rho = \frac{1}{2}$ and 1.2532 at $\rho = 0.9$, with a *vertical asymptote* at $\rho^\circ = 1 - c_d/a$ =0.980. Past that drift rate, compiling never pays at any budget, and the asymptote arrives long before the environment is fully unstable.

Third, the threshold **diverges in** c_d before the price becomes prohibitive: 2.279 at $c_d = 0.5$ and 21.73 at $c_d = 0.85$, against a limit of 0.891. A system one per cent away from the break-even price needs two orders of magnitude more budget to prefer compiling.

Fourth, and easiest to miss, a real chain completes *whole* rungs. The fluid threshold above is where the expected depth ties; the budget at which a scheduler first sees the compiled arm get further ahead is a different number, because a budget-constrained run stops just after a cheap replay and its realised count is biased above $B/\bar{c}_d$. Writing k for the number of replays affordable beyond the verification, the whole-rung threshold is the smallest k with $1 + \sum_{j \le k} p^j > (c_v + c_d + S + kc_c)/c_v$. At $S = 1$ it is 2.020 against a fluid value of 2.122, and it lies below the fluid threshold for every S that matters. The geometric sum also creates a *second* critical premium, $S^\circ = \max_k[\sum_{j \le k} p^j - kc_c/c_v]$ =8.47, above which the compiled arm never overtakes a whole-rung chain at any budget. The fluid theory has no such bound; granularity introduces one.

d) Reach against budget: Depth and reach must not be confused here, and this is where the confusion is most tempting. Reach is exponential in the saturated difficulty coordinate of (1), applied to depth as $D(K) = 1 - e^{-K/K_0}$ with K_0 =10 rungs. Therefore

$$R_{\text{unsolid}}(B) = A_0 \exp\Big[\alpha\big(1 - e^{-B/(c_v K_0)}\big)\Big], \qquad (28)$$

$$R_{\text{solid}}(B) = A_0 \exp\Big[\alpha\big(1 - e^{-K_{\text{solid}}(B)/K_0}\big)\Big], \qquad (29)$$

with $\alpha = \ln 10^{12} = 27.63$, the paper's horizon, and A_0 a normalisation. Their ratio is

$$\frac{R_{\text{solid}}}{R_{\text{unsolid}}} = \exp\Big[\alpha\big(e^{-K_A/K_0} - e^{-K_B/K_0}\big)\Big] \;\ge\; 1, \qquad (30)$$

which *returns to one* as $B \to \infty$: both arms saturate at the horizon $A_0 e^{\alpha}$, so beyond it extra depth stops buying reach. The ratio peaks at a finite $B^\dagger$ =3.71 with a value of 2.4×10^7, is 25806.9 at $B = 10$, 1.071 at $B = 60$ and 1.0002 at $B = 120$. The depth advantage, by contrast, grows without bound with slope $1/\bar{c} - 1/c_v$ =8.174 per unit budget. *The depth advantage is unbounded and the reach advantage is not*, and quoting one as the other is the error this subsection exists to prevent. Locally, in the window $B - c_v \ll c_v K_0$, the ratio grows exponentially at the rate $\alpha(1/\bar{c} - 1/c_v)e^{-c_v/K_0}/K_0 = 20.437$ per unit budget — exponential locally, bounded globally, the same shape as §II-H and for the same reason: the coordinate is convex.

e) Falsifiability: The claim that can be wrong is short, and it is not the one the section started from. It is: *there is a budget below which a system whose compiled path carries a fixed premium S should re-verify every rung rather than compile, and that budget is $B^*(S) = c_v + S\,G_\infty/(G_\infty - 1)$.* Three measurements would refute it. A fitted $B^*(S)$ that is not affine, or whose intercept is not the price of one crossing, means the premium is not fixed and the amortisation picture is missing a state variable. A ρ-dependence at $c_d = 0$ means the drift term is not multiplicative in the per-rung price. And a monotone reach ratio means the difficulty saturation is not binding in the regime tested, which would relocate the horizon rather than confirm an explosion. All three are cheap to test, and the third is the one that separates this account from the "compilation makes reach explode" story it replaces.

TABLE XIV

EXPERIMENT J. THE CRITICAL BUDGET AGAINST THE FIXED PREMIUM S: CLOSED FORM (27), THE ESTIMATE OBTAINED BY DRAWING THE PER-RUNG PRICE FROM THE STATED PROCESS AND SOLVING THE CROSSING, BOTH WITH AND WITHOUT WHOLE-RUNG GRANULARITY, AND THE DEPTH GAIN AT A FIXED BUDGET OF TEN. $c_v = 1$, $c_c = 0.01$, $\rho = 0.1$, $c_d = 0$, SO THE PREDICTED SLOPE IS $G_\infty/(G_\infty - 1)$ =1.1223AND THE INTERCEPT IS THE PRICE OF ONE CROSSING.

S	B^* closed	B^* estimated	B^* whole rungs	ΔK at $B = 10$
0.00	1.000	1.000	1.010	73.57
0.25	1.281	1.281	1.260	71.28
0.50	1.561	1.562	1.510	68.98
1.00	2.122	2.125	2.020	64.39
2.00	3.245	3.249	3.030	55.22
4.00	5.489	5.498	5.060	36.87

TABLE XV

EXPERIMENT J. B^* AGAINST THE MONITOR PRICE c_d AND THE DRIFT RATE ρ. FINITE ENTRIES ARE (27); "∞" MEANS NO BUDGET MAKES COMPILING RATIONAL, WHICH HAPPENS EXACTLY WHEN $c_d \ge (1 - \rho)(c_v - c_c)$ — THE PRICE BREAK-EVEN OF §IV-F. NOTE THAT RAISING ρ AT FIXED c_d RAISES THE THRESHOLD AND REACHES INFINITY AT 0.980, NOT AT ONE.

c_d	$\rho = 0$	$\rho = 0.1$	$\rho = 0.5$	$\rho = 0.9$
0.00	1.000	1.000	1.000	1.000
0.02	1.021	1.023	1.042	1.253
0.10	1.112	1.126	1.253	∞
0.25	1.338	1.390	2.020	∞
0.50	2.020	2.279	∞	∞
0.80	5.211	9.791	∞	∞
0.99	∞	∞	∞	∞

H. Nested solidification: reuse rate κ and the compound interest of interfaces

§IV-G ended on a deflationary note: the gain from compiling one interface is a price ratio, and a price ratio is a constant. That is the correct reading of (24), and it is why the previous subsection's advantage saturated in reach. But it is not the whole economy of a hierarchy. A system whose bottom layer is compiled does not merely replay its own rungs faster; it acquires the option of *certifying a higher layer by citing the compiled one*. If a parent's correctness follows from labels that already exist below it, then the parent is not re-proved at all, and the marginal cost of the parent stops tracking the complexity of its children. This subsection states that

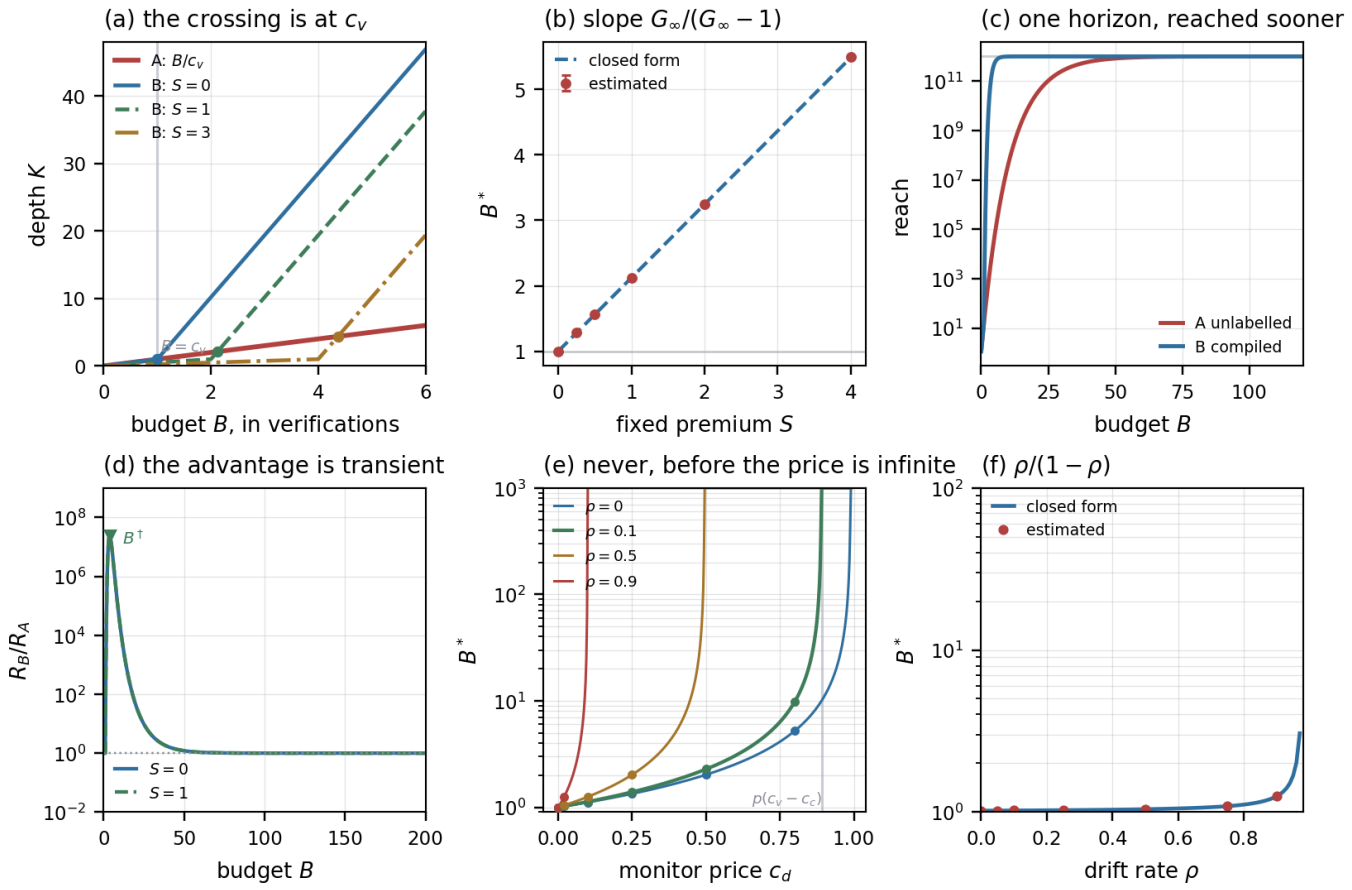


Fig. 11. Experiment J. (a) Depth against budget. The unlabelled arm is a line of slope $1/c_v$; a compiled arm with no premium hugs it and then separates, and the dots mark B^* for three premiums. Note that $S=0$ has its crossing at c_v itself. (b) B^* against S: the closed form is a line of slope $G_\infty/(G_\infty-1)$ through c_v, and the estimates sit on it. (c) Reach of both arms on a logarithmic axis: the compiled arm arrives at the horizon sooner, and both stop at it. (d) The reach ratio, with the same logarithmic axis: it dips below one inside the losing region when $S>0$, crosses at B^*, peaks at $B^\dagger$ and returns to one — the plot form of the statement that the advantage is transient in budget. (e) B^* against the monitor price at four drift rates; the curves diverge at $p(c_v-c_c)$, so the threshold becomes infinite before the price does. (f) B^* against ρ at the baseline monitor price: a hyperbola with an asymptote at 0.980. With no monitor the curve would be flat at c_v, which is why the threshold is a statement about the monitor and not about drift.

mechanism, prices it, and derives the single scalar that decides whether it compounds or merely discounts.

a) Two layers: Layer 0 is the base: the physics, the hash, whatever the certificate ultimately rests on. It costs c_v to verify a leaf, and it becomes compilable at the budget of (27) with $c_d = 0$ and a fixed premium S; at the baseline $S = 5$ that budget is 6.61, or 6.08 for a whole-rung scheduler. Layer 1 is a task family of *dimension depth* d: each level composes $L = 4$ objects from the level below. The two arms differ only in what a higher level is allowed to say about a lower one.

The un-nested arm verifies a top-level task by re-verifying every leaf it touches:

$$c_{\text{flat}}(d) \;=\; c_v\, L^d, \tag{31}$$

exponential in the number of dimensions, and the honest price of complexity when nothing is shared. The nested arm, once layer 0 is solid, replaces the verification of a child by a *consultation* of that child's compiled label. The replacement that the whole subsection rests on is

$$c_{\text{top}} \;=\; \text{lookup}(L_0) \;=\; \bar{c} \qquad \text{rather than} \qquad c_v(L_1), \tag{32}$$

with $\bar{c} = 0.109$ the drift-adjusted replay price of (23). Equation (32) is not an identity of nature; it is the claim that a compiled object can be cited instead of re-derived, and it is exactly the claim the falsifiability paragraph below puts at risk.

b) The recursion, and where it stops depending on d: A consult succeeds with probability $\kappa \in [0,1]$, the *reuse rate*: the fraction of a child's verification that the parent's label makes redundant. If it succeeds the consult costs $\bar{c}$, at any depth; if it fails the child must be traversed, which costs whatever a task one level down costs. Writing m_d for the marginal price of one more task at depth d,

$$m_0 = \bar{c}, \qquad m_j = \kappa\,\bar{c} + (1-\kappa)\,L\,m_{j-1}. \tag{33}$$

At $\kappa = 1$ the recursion collapses: $m_d = \bar{c}$ for every d, so *the price of a top-level task no longer depends on how many*

dimensions it spans. At $\kappa = 0$ it inflates to $m_d = \bar{c}\, L^d$, which is (31) multiplied by the constant $\bar{c}/c_v$: amortisation with no reuse, and no compounding whatsoever. The multiple of flat tasks bought by one nested task is

$$M(d) \;=\; \frac{c_v\, L^d}{m_d}, \tag{34}$$

and since $u_d \equiv m_d/\bar{c}$ obeys $u_d = \kappa + (1-\kappa)L\, u_{d-1}$, the multiple grows geometrically in d with base

$$\min\Big(L,\; \frac{1}{1-\kappa}\Big). \tag{35}$$

Three readings of (35) are the substance of this subsection. First, at $\kappa = 0$ the base is 1: $M(d) = G_\infty = 9.174$ at every depth, a constant, which is precisely the objection that amortisation is arithmetic. Second, at $\kappa = 1$ the base is L and $M(d) = G_\infty L^d$. Third, and least obvious, the base *saturates*: increasing κ beyond

$$\kappa^* \;=\; 1 - \frac{1}{L} \;=\; 0.75 \tag{36}$$

buys nothing, because m_d is already bounded in d iff $(1-\kappa)L < 1$. The marginal cost is bounded above the knee, grows only linearly *at* it, and grows geometrically below it. Compounding in the dimension is therefore not a property of compiling at all; it is a property of how much of a child a parent's label is allowed to replace.

c) Two thresholds, and a valley between them: The budget axis does not collapse to a single number. The base becomes compilable at B^*, but the higher levels cost S each to relabel and deliver nothing until the whole chain is mapped, so the one-time cost of a depth-d capability is

$$T(d) \;=\; B^* + \sum_{j=1}^{d} (S + L\, m_{j-1}) \;=\; B^* + d\,(S + L\,\bar{c}) \tag{37}$$

for $\kappa = 1$, linear in depth while (31) is exponential in it. One task is delivered after $T(d)$, and each thereafter at m_d, so the two arms tie at

$$B_{\text{rec}}(d) \;=\; \frac{T(d)}{1 - m_d/c_{\text{flat}}(d)}, \tag{38}$$

which sits just above $T(d)$ whenever nesting is worth doing. Between B^* and B_{rec} the nested arm is *strictly behind*, and this is the first non-degenerate loss region in the paper: §IV-G had to conclude that the compiled arm is never behind anywhere, because its premium was paid once. Here the premium is paid per level, before any level delivers, and the valley is real. At the baseline, $T(6) = B_{\text{rec}}(6) = 39.23$ for a six-dimensional task: the arm spends 32.6 further units of budget, after the base is already compiled, producing nothing measurable.

Table XVI walks a budget across both thresholds. Below B^* the nested arm *is* the flat arm, because nothing has been compiled and no premium has been paid; at B^* it falls to zero output while the flat arm keeps grinding; at B_{rec} it re-emerges at parity; and from there the ratio climbs to 30 207 at $B = 200$. The marginal price is a step, from 4096 to 0.109, a factor of 3.76×10^4 at $d = 6$ and 6.01×10^5 at $d = 8$ — and the step is in the *dimension* coordinate d, not in the budget coordinate B: on the budget axis the same fact appears as a kink.

TABLE XVI

EXPERIMENT K. THE LOSS REGION AND THE TAKE-OFF. BOTH ARMS COMPLETE TOP-LEVEL TASKS OF DIMENSION DEPTH $d = 6$ WITH BRANCHING $L = 4$ AGAINST A BUDGET B, AT $c_v = 1$, $c_c = 0.01$, $\rho = 0.1$, $S = 5$, $\kappa = 1$. THE NESTED ARM FALLS TO ZERO AT $B^* = 6.61$ — NOT BECAUSE COMPILING IS DEAR, BUT BECAUSE THE PER-LEVEL PREMIUM IS BEING PAID AND NOTHING HAS BEEN DELIVERED — AND BREAKS EVEN AT $B_{\text{REC}} = 39.23$. THE FINAL COLUMN IS THE RATIO OF COMPLETED TASKS; "–" IS ZERO.

B	N_{flat}	N_{nested}	$N_{\text{nested}}/N_{\text{flat}}$	regime
5.00	0.00122	0.0012	1	flat
6.61	0.00161	0.0000	–	B^*: investing
20.00	0.00488	0.0000	–	no output
38.73	0.00945	0.0000	–	no output
39.23	0.00958	0.0096	1	B_{rec}: break-even
40.00	0.00977	7.0856	726	compounding
60.00	0.01465	190.5718	13010	compounding
100.00	0.02441	557.5443	22837	compounding
200.00	0.04883	1474.9755	30207	compounding

d) Falsifiability: The claim is not that compiling is cheap, and it is not that hierarchies always compound. It is: *nesting beats independent verification only if the measured reuse rate exceeds $\kappa^* = 1 - 1/L = 0.75$, and below that rate the system should stop adding levels.* The criterion is a single scalar, measurable without knowing any price: instrument a parent's verification, count the fraction of child checks that its label discharges, and compare with $1 - 1/L$. Three measurements would refute the account. A system with $\kappa > \kappa^*$ whose marginal cost still grows with the dimension count falsifies (33), which would mean the label is not being consulted at the level the parent composes at. A system with $\kappa = 0$ whose cost nevertheless compounds falsifies (35), and would mean the amplifier is something other than reuse. And a fitted $B_{\text{rec}}(d)$ that is not linear in d means the per-level premium is not fixed — the same failure mode, in the budget coordinate, as a non-affine $B^*(S)$ in §IV-G. Each test is cheap, and the second is the one that separates this account from the "hierarchies are exponentially efficient" story it replaces.

e) What the compounding is not: Three limits must be stated with the result, or the section will be read as claiming more than it shows. First, (34) compounds in *cost* at fixed prices; it says nothing about time, and nothing about value. If the value of a top-level task saturates in d the way reach saturates in depth — which (30) already forces at the horizon — then the multiple in value is capped even where the multiple in cost is not, and the two must not be quoted as one. Second, κ is an assumption in this experiment and a measurand in the next; the simulation checks that the tree walk prices the recursion it claims to price, which is a consistency result, not evidence that any compiler discharges three quarters of its children's checks. Third, the loss region is not a technicality: a system that buys a hierarchy and never runs enough tasks to amortise (37) is *worse* than one that never built it, and B_{rec} says by how much budget. High reuse makes the upper layer

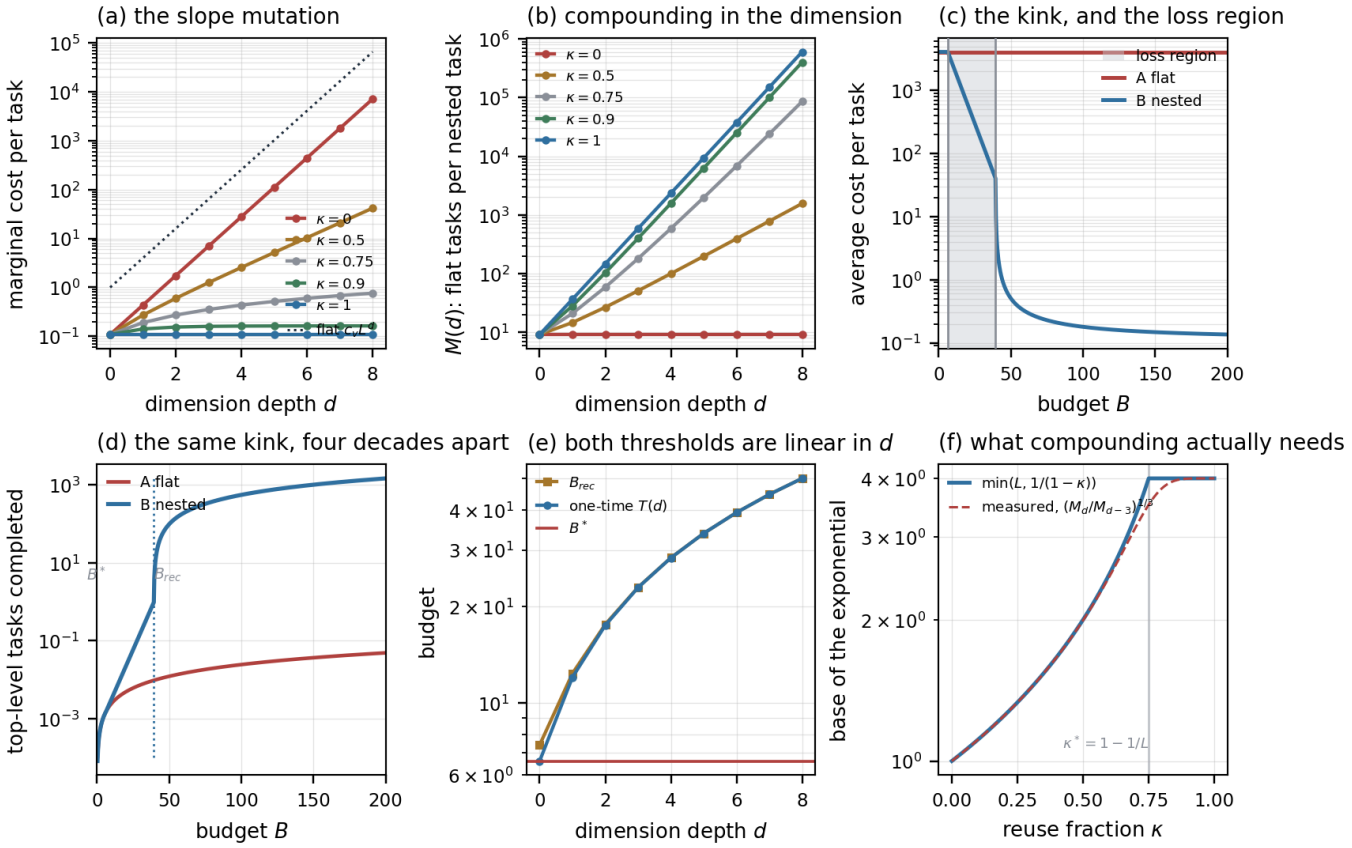


Fig. 12. Experiment K. (a) Marginal cost of one task against dimension depth d at five reuse rates. The flat arm is $c_v L^d$; $\kappa = 0$ tracks it exactly up to the constant $\bar{c}$, which is what amortisation alone can do. At $\kappa = 1$ the line is horizontal: the price of a task stops depending on its dimension count. (b) The compound multiple $M(d)$. It is flat at G_∞ for $\kappa = 0$ and grows with base L for $\kappa \geq \kappa^*$. (c) Average cost per task against budget, with the loss region shaded: the nested arm is undefined there by construction, since it is paying for levels it has not finished. (d) Completed tasks against budget on a logarithmic axis, the same kink four decades wide. (e) Both thresholds against depth: linear in d, because the relabelling is linear even where the task is not. (f) The compound base against κ: $\min(L, 1/(1-\kappa))$, with the knee at $\kappa^* = 1 - 1/L$ marked. The measured values are $(M_d/M_{d-3})^{1/3}$ from panel (b), a ratio that cancels G_∞. They sit on the prediction away from the knee; *at* the knee they sit below it, by the finite-window factor $(u_{d-3}/u_d)^{1/3}$ with $u_d = 1 + \kappa d$ — a factor that closes only as $d \to \infty$, and not a fitted offset.

eat the lower layer's complexity; it does not make the purchase free.

I. Is AI OS the next crossing?

The model makes a specific claim about what comes after a crossing: the new gap is inside the space just connected, it is strictly narrower, and it inherits the reach. After biology crossed the quantum gap, the next gap is not "more quantum coherence"; it is *how to build the structure instead of growing it*—the biomimetic-materials problem. Evolution had a billion years and a population to search with; a laboratory has neither, which is why the successor difficulty is higher even though the physics is easier to describe.

The same structure applies to agent systems. For a semantic layer, the narrow condition is roughly "a machine can hold and re-check a claim across a context boundary", and the quantity that governs reach is not raw model capability but *witness density*: the fraction of claims that arrive with an artefact a third party could check. Under (6), a system with high fidelity and a high-density witness trail has a small exponent and therefore large reach, while a system with better benchmarks but a sparse trail has a large exponent and stays narrow. This remains a conjecture: §III-G tests the weaker claim that a scaling law can be *read* as a fidelity curve, and finds the reading consistent but unable to discriminate; it says nothing about witness density. It is stated because it is falsifiable—measure witness density and reach independently in deployed systems, and the model predicts they covary with the exponent of (6) rather than with capability.

J. Threats to validity

1) **The experiments validate a model, not nature.** The power law is derived; the simulation confirms that the implementation realises it and locates where it stops. The physical content rests on four literature anchors, and on one assumed reach anchor which we swept.
2) **The inverse-rarity assumption is an assumption.** Equation (3) is a modelling choice with a plausible story, not a thermodynamic result. A different rarity model would change (6).

3) **The room-temperature coherence claim is contested.** We encode the sceptical reading [18]. Adopting the optimistic reading [17] would strengthen the biological reach result and weaken our argument that reach is independent of coherence—so the sceptical choice is the conservative one for our thesis.
4) **Slack noise is assumed proportional to remaining room.** An absolute σ would be unphysical at high fidelity; the proportionality constant is not measured.
5) **The span horizon censors 25.6% of crossings.** Those rows are excluded from regression rather than clipped, which biases nothing but discards a quarter of the sample.
6) **One gap at a time.** The model has no networks, no shared conditions, and no cost coupling between crossings. A system of interacting gaps is future work.
7) **Three of the four boundary fidelities are assumptions.** Experiment D rests on a table in which one number is measured. The ordering of the four boundaries is robust to moving any assumption by a factor of two; the statement that *no* boundary is seamless is not, because the quantum boundary is 0.30 nines from the threshold. We report the margin for that reason.
8) **The noise model is linear and complementary.** It assumes a boundary passes the condition with probability F and everything else with probability $1 - F$. A real filter's pass and stop bands are not exact complements, and the cryostat/scaffold contrast is precisely where they diverge—a cryostat rejects noise by rejecting nearly everything, including its own support infrastructure. The *ratios* in Table VIII survive that objection; the absolute over-budget factors do not.
9) **Selectivity is not boundary fidelity.** The two are distinguished in §II-G and §III-F, but the engine's single "amplification" field still mixes them with multiplicity. Experiment E works in the law's own units for that reason, and compares exponents rather than absolute reach.
10) **Experiment F rests on an untested bridge.** Equation (17) converts a per-token loss into a boundary fidelity, and no scaling-law measurement can validate that conversion: a systematic error in it moves every number in §III-G while leaving every fit statistic unchanged. What the experiment does establish does not depend on the bridge's *form*, only on F being a per-boundary probability composed multiplicatively—which is what the depth-width prediction tests.
11) **The emergence section inherits the scaling law's own disputes.** The two laws carried differ in exponent by 4.5; we report both rather than choosing, but a third law would give a third k.
12) **A hash witnesses a state, not an action.** This is the sharpest limit in §IV-D and no implementation removes it. An agent that finds the intended artefact already in place and does nothing produces a witness matching the intent, and the two histories are indistinguishable because they *are* the same final state. The measured zero false-completion rate is therefore a property of a model in which every failure changes the state; a failure that preserves it is unobservable in principle rather than merely unobserved.
13) **The witness is only as strong as its registration.** The intended digest has to arrive from outside the run. Poison that supply and the membrane certifies whatever is on disk, because a hash compared against an expectation derived from itself is the tautology it appears to be. A guarantee that holds only for honest registration is a guarantee about the registrar, and the mechanism by which a registrar is kept honest is not something this paper supplies.
14) **The weak baseline is a keyword scan, deliberately.** §IV-D compares the causal judge against a rule that looks for error strings, not against a language model reading the log. A stronger reader would move 79.8% but not the ceiling, which is set by what the harness emits; the choice is made so that the causal result cannot be an artefact of a weak opponent. It is not evidence that any particular LLM judge is as poor as the scan.
15) **One failure mechanism per task.** The generative model draws a single failure mode, so the measured channels do not overlap and their contributions add. A system that fails twice, once visibly and once silently, would show a lower causal sensitivity than reported here, and by an amount this experiment does not bound.

V. Conclusion

An interface is not a published surface. It is a narrowing condition whose satisfaction buys reach, and the price of the crossing is set by the mechanism rather than by the gap. Formalising this yields a law with a testable shape: $\mathrm{Reach} \propto (1 - D)^{-k}$ with $k = \gamma P_0 |\ln s|$, a family indexed by fidelity rather than a single curve.

Across 7000 simulated crossings, the family is recovered to within 0.4% and the single curve is rejected at $R^2 = 0.12$. The counterfactual is the sharpest result: a mechanism that extends coherence by nine orders of magnitude gains two orders of reach, while a mechanism that extends it by nothing gains five, because replicability—not coherence—sets the scale. The failure boundary is the mechanism's own fidelity, and the fractal ladder terminates at a representational ceiling of $1 - 10^{-9}$, 24 generations from $D = 0.99$.

Two further results are not fits but small theorems, and they explain why the numbers above look the way they do. Seamlessness is a threshold on resistance, so it arrives in whole nines: a boundary at 1.9 nines is not nearly transparent but not transparent, and a five-stratum stack cannot be made seamless by improving one boundary, because depth costs $\log_{10} n$ extra nines. And the exponential form of the law is a local truth—exact in the narrowing coordinate and along the ladder, tangent in difficulty with a rate $k/(1-D)$ that diverges, and impossible globally with a constant rate because $\ln \mathrm{Reach}$ is strictly convex. Reach does grow explosively near $D = 1$.

What a single α describes is the tangent, and a paper that reports one has measured the curve's slope at a point without measuring its curvature.

The same law was then applied to a language-model scaling law, and it is the one place where the model was turned on a claim from outside its own domain. The law survived and the explanation did not. Reach follows the power law exactly, but the fit discriminates nothing—α and P_0 are the only free quantities and the law fixes their product—so the falsifiable content is the implied selectivity, $s = 0.46$. The critical scale near 20B is real and reproducible, but at it the boundary is at 0.92 nines rather than the two nines the hypothesis requires; two nines lie 1771 times further out. What the model does contribute is a reason for the step's *appearance*: composing boundaries translates the accuracy curve without steepening it, so every task depth shows the same 3.94-decade transition, and sampling model sizes a decade apart puts four points inside it. That is a claim with a number, and it fails if a deeper task shows a narrower transition.

The last experiment turns the same instrument on the paper's own remedy. A witnessed system does not convert depth into an explosion, and cannot: completion of a ladder is bounded by completion of its prefixes, so reach is non-increasing in depth whatever the per-layer law is. What the label buys is not reach but the absence of an invented one—at $D = 0.99$ and depth 6 the transcript overstates the causal rate by 2.08×10^{10}, while the label's genuine gain is 9.3 orders of magnitude in the *cost of depth*. A discount belongs at the margin, where it can be spent; spent on the exponent it would be a different claim, and one the arithmetic does not permit.

That discount, priced, splits in two. On a single interface it is arithmetic: a compiled traverse costs $\bar{c}$ instead of c_v, and no budget turns a price ratio into an exponent. Across layers it is not arithmetic, because a compiled layer can be *cited* by the layer above it, and citation is what removes the dimension count from the price. Writing κ for the reuse rate—the fraction of a child's verification its parent's label discharges—the marginal cost of a task at depth d obeys $m_d = \kappa\bar{c} + (1-\kappa)Lm_{d-1}$, so the advantage over independent verification grows with base $\min(L, 1/(1-\kappa))$: exactly G_∞ at $\kappa = 0$, and L for any $\kappa \geq \kappa^* = 1 - 1/L$. A system below the knee compounds nothing however much it compiles, and a system above it gains nothing further by compiling harder. Economically this is the content of seamlessness: high reuse lets the upper layer's marginal cost swallow the lower layer's complexity, which is why a transparent interface is worth building and a merely fast one is not. The budget that buys that position is not B^* alone; the relabelling is paid per level, so the arm is strictly behind between B^* and $B_{\text{rec}}(d) = T(d)/[1 - m_d/c_{\text{flat}}(d)]$, and a hierarchy bought and never exercised is worse than none.

The honest summary is narrower than the ambition. What has been shown is that a particular formalisation of interfaces is internally consistent, that its central law is recovered numerically to within a fraction of a percent, that its exponent is mechanism-specific, and that it breaks in two places that can be located exactly. What has not been shown is that the physical anchors are right, or that inverse rarity is the correct model of reach. Both are stated as assumptions above, and both are the experiments we would run next.

APPENDIX A
SELF-AUDIT OF THIS PAPER

A package built to refuse unwitnessed claims should not publish a manuscript that never turns that refusal on itself. So the audit was run on the paper. The instrument is `experiments/paper_self_audit.py`: it reads this file, locates each claim by a pattern that must match exactly once, re-derives the numbers printed above from the modules that generated them, and writes the table and counts below. If an anchor moves, or a module and its macro disagree, the audit fails instead of quietly reporting on a paper it did not read. It reports 12 claims: 7 PASS, 2 INVALID, 2 UNAVAILABLE, 1 BLIND, and 0 escalated to failure.

a) What each verdict is allowed to mean: The four values are the package's own, and their scope is quoted rather than paraphrased. PASS in Table XVII is mechanical: the printed number and the module that produced it agree at the printed precision, and for the knee the check re-tests the property as well as the value, because a number that bounds nothing would still print 0.75. It does not mean the model is right about the world; it means a reader who checks the arithmetic will find it holding. INVALID means no witness was offered—*nothing to examine*, which is a different finding from *refuted* and is kept apart from it here for the same reason the engine keeps it apart everywhere else. UNAVAILABLE means the instrument that could answer does not exist in this setting, and an environment that cannot answer must not be allowed to accuse. BLIND is a statement about the registry's coverage, not about nature.

b) The two premises we cannot test: The reading of an interface as a boundary, and of causation as labelling, are the axioms the model is built on. No experiment inside this paper could decide either, and a paper that claimed to have decided one would be the exact artefact the rest of this work refuses. Both are recorded as UNAVAILABLE: *This is a philosophical premise, not a falsifiable hypothesis.* They are printed here rather than buried because a reader is entitled to know which of the paper's load-bearing statements are definitions, and because a premise that is invisible is one that gets cited as a result.

c) The two assumptions we have not measured: The per-layer success probability is set to $p_{\text{layer}} = 1 - D$. That is a choice, it is stated as one in Section III-G, and every ordering reported there survives any monotone decreasing replacement; but no measurement of a per-layer success rate exists at any of the five scales, so the row is INVALID—unwitnessed, not wrong. The conjecture that reach tracks witness density rather than capability is likewise INVALID: it is falsifiable and has not been falsified, which is a different and better position than the two premises above, and the difference is the reason the values are kept distinct.

d) The partition we cannot see past: Scale is partitioned into 5 strata joined by 4 boundaries. Of the 10 unordered pairs this admits, 6 have no filing and therefore no witness, and whether a sixth stratum exists is a question the registry cannot pose at all. That is a gap in the taxonomy, not evidence that the missing boundaries are uncrossable, and it is why the row is BLIND rather than a failure.

TABLE XVII
SELF-AUDIT OF THIS PAPER. EACH CLAIM WAS GRADED WITH THE SAME LATTICE THE PAPER APPLIES TO OTHER PEOPLE'S FILINGS, AGAINST THE EVIDENCE CHANNEL IN THE LAST COLUMN. A PASS MEANS THE MANUSCRIPT'S ARITHMETIC RECOMPUTES, NOT THAT THE CLAIM IS TRUE OF NATURE; AN INVALID MEANS NO WITNESS WAS OFFERED, NOT THAT THE CLAIM IS FALSE.

id	claim	verdict	evidence channel
A1	The critical-budget closed form	PASS	recomputation from experiments/critical_budget_experiment.py
A2	The amortisation ceiling	PASS	recomputation, printed by two independent generators
A3	The slope nothing was fitted to	PASS	recomputation from the two prices
A4	The reuse knee, with bounded marginal cost above it	PASS	recomputation plus a two-sided boundedness test
A5	The recovery budget	PASS	recomputation, plus the inequality B_rec > T
A6	The efficiency jump at B=200	PASS	recomputation, and a read-back of the generated table
A7	The reach advantage peaks inside and returns to one	PASS	numerical re-solution, no generated constant involved
B1	Causation is labelling: the do-operator separates the two probabilities	UNAVAILABLE	no instrument: the do-operator is estimated inside a simulated judge, not in a physical system
B2	An interface is a boundary, not a surface	UNAVAILABLE	definitional: no measurement could decide it
C1	The per-layer success probability is 1 minus the difficulty	INVALID	no witness: no measurement of a per-layer success rate at any of the five scales
C2	Reach tracks witness density, not capability	INVALID	no witness: stated as a conjecture, no deployed system measured
D1	Scale partitions into the five strata	BLIND	registry statement: coverage of the strata graph, not of nature

e) What the exercise cost: One audit implementation bug, found by the audit itself: the first version tested the knee's claim by checking that the marginal cost was "large" below κ^*, which failed on a true statement because growth is geometric, not explosive. The test now checks the recursion identity $(m_d - \kappa\bar{c})/m_{d-1} = (1-\kappa)L$ exactly. The episode is reported because an auditor whose own errors are invisible has no standing to report anyone else's, and because the corrected check is the one printed above.

A. Auditing the Bibliography

The claims above are the paper's arguments. Its citations are claims of a different kind—*this identifier names this paper*—and they are the ones an author is most able to get wrong by ac-

cident and most tempted to leave unchecked. We ran our own `CitationAuditProxy` against the `references.bib` of this paper: each entry's DOI is resolved against CrossRef and each arXiv id against the arXiv API, and the record's returned title is compared with the title the entry claims. That comparison is the witness. Nothing about the manuscript is consulted.

Of the 29 entries, 27 are PASS: an agency answered and the title matched (16 through CrossRef, 11 through arXiv), and 2 could not be witnessed at all. The FAIL count is 0, the verdict that would matter most: a right-looking identifier that resolves to a different paper is the fabrication this audit exists to catch, and none was found—a statement about this bibliography and not about bibliographies in general. Both INVALID entries are genuinely unwitnessable: a 1982 book, and a 1962 print article from a journal volume that predates DOI registration. No agency can ever speak for either, so deleting them to make the table green would be the same false completion the rest of this work refuses. They are kept, and their existence is recorded as a *locator*—a catalogue record for the book, a scanned volume index for the article—so that the reader can go and look rather than take our word. A locator is not a witness, and Table XVIII keeps the two apart.

a) The first run was the finding: On its first run the auditor returned 0 PASS and 19 INVALID, with no FAIL. Not one entry carried a DOI or an arXiv id: the bibliography was a set of assertions that nothing outside itself could check, which is precisely the shape of claim this package exists to distrust. The verdict was not that the citations were false but that they were unwitnessable, and the repair was to give each one the identifier under which it could be re-resolved. The zero FAIL in that run is not praise: with no identifier present there was nothing to contradict. The pre-audit file is kept as `data/references_pre_audit.bib` so that the claim can be re-run rather than believed.

b) How the identifiers were chosen: By title search, not by memory, and the distinction earned its keep: for most entries the first hit returned by the agency was a *different* paper with a similar title—a different spectroscopy review, a different complexity essay, a 2024 paper about proteins rather than language models. Each candidate was accepted only after the auditor's own title comparison passed. A search that returns a plausible DOI is not a witness; a record whose title is the entry's title is.

TABLE XVIII
BIBLIOGRAPHY AUDIT. THE VERDICT COLUMN IS THE AUDITOR'S OWN; THE LAST COLUMN DISTINGUISHES AN ENTRY AN AGENCY WITNESSED FROM ONE A HUMAN LOCATED. A PASS MEANS THE IDENTIFIER RESOLVED AND THE RETURNED TITLE MATCHED, NOT THAT THE PAPER IS CORRECT.

entry	identifier	verdict	how it was witnessed
engel2007	10.1038/nature05678	PASS	crossref (title match)
brixner2005	10.1038/nature03429	PASS	crossref (title match)
panitchayangkoon 2010	10.1073/pnas. 1005484107	PASS	crossref (title match)
duan2017	10.1073/pnas. 1702261114	PASS	crossref (title match)
adolphs2006	10.1529/biophysj. 105.079483	PASS	crossref (title match)
mohseni2008	10.1063/1.3002335	PASS	crossref (title match)
plenio2008	10.1088/1367-2630/ 10/11/113019	PASS	crossref (title match)
lambert2013	10.1038/nphys2474	PASS	crossref (title match)
romero2017	10.1038/nature22012	PASS	crossref (title match)
kjaergaard2020	10.1146/annurev- conmatphys-031119- 050605	PASS	crossref (title match)
mandelbrot1982	–	INVALID	no agency; located at OpenLibrary work OL1808391W
simon1962	–	INVALID	no agency; located at archive.org scan of Proc. Am. Phil. Soc. Vol. 106 (1962), index p. 467
west1997	10.1126/science.276. 5309.122	PASS	crossref (title match)
barabasi1999	10.1126/science.286. 5439.509	PASS	crossref (title match)
clauset2009	10.1137/070710111	PASS	crossref (title match)
hoffmann2022	2203.15556	PASS	arxiv (title match)
kaplan2020	2001.08361	PASS	arxiv (title match)
schaeffer2023	2304.15004	PASS	arxiv (title match)
besiroglu2024	2404.10102	PASS	arxiv (title match)
pearl2009	10.1017/CBO 9780511803161	PASS	crossref (title match)
necula1997	10.1145/263699. 263712	PASS	crossref (title match)
wei2022	2206.07682	PASS	arxiv (title match)
greenblatt2023	2312.06942	PASS	arxiv (title match)
vanderweij2024	2406.07358	PASS	arxiv (title match)
du2024	2403.15796	PASS	arxiv (title match)
zhao2025	2502.17356	PASS	arxiv (title match)
alabdulmohsin2025	2502.14924	PASS	arxiv (title match)
torkamandi2025	2501.04286	PASS	arxiv (title match)
gill2026	10.2139/ssrn. 7307578	PASS	crossref (title match)

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